Svach

IB MYP Design E-Portfolio 2024-2025

Designing For Well-Being



Key Concept: Development
Related Concept: Adaptation
Global Context: Scientific and Technical Innovation

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Criterion A - Inquiring and Analyzing

Strand (i) Explain and justify the need for a solution to the problem.

Well-being is something which doesn't just constitute ‘the state of being happy, healthy, prosperous, and comfortable’, but also a condition, and a factor, which sustains the very essence of the contemporary life we all live. With developments in effectively every domain around us, urging us to **adapt** to our incredibly volatile and uncertain surroundings, the **healthy practices** which we employ will have a significant impact on developing and shaping this very sustenance of life, which we call well-being. While well-being can be divided into numerous constituent aspects, the major ones include physical exercise, mental strength and tranquility, quality sleep, strong relationships, balanced screen time usage, and nutritious diets, as mentioned by the ‘BetterHealth Channel’ of the Victoria State Government’s Department of Health.

Furthermore, the practices associated with the maintenance of one’s personal well-being are much more than determinants of one’s health and wellness, they are major antecedents of workplace productivity, academic success, and overall life satisfaction and longevity, having profound, indelible, impacts on an abundance of the components of one’s personal and professional condition. This thereby underscores the fact that in order to keep up with the challenges and obstacles posed by the volatile, uncertain, complex, and ambiguous (**VUCA**) future that approaches, ensuring that one inculcates the above constituent aspects of well-being positively and effectively into their daily lives is imperative, and failing to do so can result in significant hurdles and problems in relation to practising a healthy lifestyle.



Fig: Mindmap of Well-Being Subdomains

In today’s world, the vast array of **scientific and technical innovations**, which have uplifted and advanced the human race from what it was a few centuries ago, undoubtedly have numerous unintended consequences, ultimately having the potential to have serious repercussions on the aforementioned aspects of well-being. With the issues of prolonged usage of electronic devices and screen-time management coming more into light, along with the condition of pediatric obesity and physical exercise routines for juveniles experiencing greater developments, accompanied with a vast amount of research and statistics on sleep hours and tracking, one area which is often ignored and doesn’t experience much innovation, especially for children and school students, is **nutrition**. While a plethora of solutions are already conceptualized to combat issues with child nutritional deficiencies, such as supplements and parenting techniques, the underlying problem lies within the inability to effectively implement and actively follow a healthy and balanced diet, that complements one’s physical exercise, is personalised to their interests and dietary preferences, and most importantly, contains palatable and satisfying food.

This problem isn’t just a mere fact, it is supported by concrete data, statistics, research, and studies. In a survey with Middle School and High School students in California, 45% of students mentioned that they disliked the school meals, while 34% stated that they skip meals because they dislike the food being served, demonstrating the sheer dissatisfaction of a large number of students with their school meals and nutrition. These results further indicate that these school meals, which comprise a vast majority of a student’s meals in a week as well as their required nutritional elements, often don’t match the interests or preferences of students, lack quality, and most importantly, aren’t palatable. Additionally, the same survey provided results that 64% of the students did express interest in eating school meals provided there was a greater amount of interesting choices, thereby showing the overall demand of individuals, which is for personalized meals that match their interests.

While this problem seems to have an easy solution, there is more than what appears to the eye. Even though ideas such as online recipes and meal planners do come to mind, as well as their vast offerings and personalization aspects which allow users to filter out various features that interest them, they are rarely available, difficult to find, and are incredibly expensive, resulting in limited usability. Although these diet programs do provide consumers with personalized dietary recommendations, they are often highly expensive, requiring substantial payments for consultations as well as premium products, making them inaccessible. Furthermore, they don't take into mind the kind of ingredients the users have access to, are not **adaptive** to the users' dietary preferences and dynamics, and overall provide options which are **more focused on what they have to offer, rather than what the users truly require**. This leaves a summarised problem statement that **most diets and meal planners aren't able to offer simultaneous personalisation, palatability, and nutritional value to teenage school students, often lacking a sense of simplicity, adaptiveness, accessibility, and cost-effectiveness.**

In order to move forward with investigating my identified problem statement, having a fixed target audience will be extremely beneficial to ensure that the entire design process is mindful of the end user, allowing me to make conscious decisions keeping in mind the specifications of my consumer group to improve the quality of my research and development. My **target audience** will comprise **teenage school students aged 14 to 18.** The reason I chose this specific age of 14 to 18 to focus on mainly surrounded the facts that they are the most affected by a variety of the well-being factors. Out of all the pupils pursuing primary and secondary education, students between the age range of 14 to 18 tend to obtain lesser hours of sleep, spend a greater amount of time on screens, and undergo higher levels of stress, mainly due to their high school academic commitments which are highly-demanding and pressurising. Furthermore, at this age, most individuals tend to have a much more developed body composition as well as a sense of maturity, which gives them the ability to perform exercises and workouts at gymnasiums to work on their physical health. Despite the fact that there is a greater focus on teenagers' health being put into such components, the aspect of nutrition tends to get overlooked, having adverse effects on their overall well-being.

Numerous common scientific and logical examples and evidences demonstrate that despite practicing various aspects of well-being such as physical health, mental health, and screen time management, various teenagers are still living unhealthy lives due to their improper and unbalanced diets, effectively pointing to the fact that nutrition has the ability to **make or break one's well-being** and **development**, but is often overlooked. Today's teenagers are incredibly health conscious, they put in considerable efforts to opt for a series of healthy diet and meal options, but are often limited because they aren't tasty most of the time, which often results in them disliking their food and daily meals. This brings up the importance of guaranteeing that the diets of teenagers do have a balance of all nutrients while being palatable at the same time, in order to complement all the other aspects of their well-being and personal satisfaction, as well as nurture the desire they have for pursuing healthy diet options by making them more palatable. Furthermore, this justifies the significance of the problem, and demonstrates the integral need for developing a solution which effectively addresses the requirements of both the client and the target audience group, ensuring they are able to consume nutritious, personalized, balanced, accessible, and palatable diets, being aware of their nutritional value and significance, and ultimately accounting for an aspect of well-being which hasn't seen much light.

The process of implementing a digital solution for this problem will also be guided by its extreme relevance to various concepts, contexts, as well as statements of inquiry. Firstly, the process will allow to make profound connections with the key concept of development and the related concept of adaptation, with all of them allowing to further realize the significance of the selected problem, acting to instill a sense of responsibility to ensure the process is culminated with an innovative and novel solution, all while exhibiting the qualities of Service as Action. By examining the problem of the lack personalised, accessible, and palatable nutritional solutions for students aged 14-18, the investigation will also allow to gain a better understanding and have a positive influence on the overall process of nutrition, which will in turn have significant impacts on the overall **development** of the teenagers in the target audience, something which is imperative as this stage, where they face numerous physical as well as hormonal changes. Furthermore, the examination of the problem has allowed to reach the understanding that the underlying issue surrounds the fact that nutrition needs to be able to **adapt** to the unique, personal, as well as volatile requirements of consumers, an integral hotspot area that requires exploration and advancement.

Exploring these fundamental concepts of development and adaptation will further permit the investigation to not just yield a helpful digital solution, but also an immense amount of learnings and reflections which can be further applied to make several transdisciplinary connections as well as in future scenarios in the IB Diploma Programs. Furthermore, this investigation will also foster deep connections with the global context of scientific and technical innovation as well as the statement of inquiry, which states 'Healthy practices can shape well-being. This is mainly sourced from the fact that its eventual aim involves developing a digital solution based on **healthy practices**, specifically the **well-being** and **nutrition** of teenage students, essentially allowing to explore **the impact of scientific and technological advances on communities and environments** and **how humans use their understanding of scientific principles**. Key examples of these scientific principles include topics such as examining the dietary requirements as well as the physiology of teenagers for the effective development of relevant nutritional and palatable diets, while the topics under technological principles involve exploring personalised recommendations and accessibility, mainly surrounding data-driven **adaptable** approaches. Overall, through this exploration these scientific and technological concepts and innovations and the manners through which they are implemented will eventually create opportunities for the betterment of society. Hence, it can be concluded that the chosen problem statement and area of study investigate several explorations associated with the unit of inquiry and global context, further taking advantage of profound connections with the key and related concepts, to establish a **highly relevant** link to Scientific and Technical Innovation.

Refer Appendices A and B for Relevant Primary and Secondary Research Justification

Strand (ii) State and prioritize the main points of research needed to develop a solution to the problem.

Research Question	Primary Sources	Secondary Sources	Relevance to Problem Statement	Priority
Describe how teenage school students perceive the diets that they consume and access. [Factual]	Grade 9-10 School Student Survey Grade 9-10 School Student Interviews	Appendix A.8	This investigation will allow to measure the extent of the problem, as well as obtain the direct opinions of the target audience on it. As a result, raw, unfiltered, as well as highly informative data can be collected on the quality of diets of school students, in order to appropriately measure the scale of the problem. Furthermore, it will provide an understanding of ideas that can be implemented to refine the standard of these diets, allowing to positively shape the students' **development** and **well-being**.	High
Identify the factors that limit teenage school students from utilizing recommendations from online diet planners and articles. [Factual]		Appendix A.20	This investigation will help understand the factors which make teenage school students online diet planners and recommendation articles ineffective for teenage school students, as well as allowing to uncover **where** the problem mainly arises from. Subsequently, an effective solution can be later developed by addressing these factors, thereby overcoming the problem.	High
How can technology be implemented in order to personalise the diets for teenage school students? [Conceptual]	Workshop on AI for Societal Benefit	Appendix A.21	This investigation will help understand the role of emerging data-driven technology and analytics in catering to personalised diets for teenage school students. Consequently, significant insights can be gained with respect to present limitations associated with the problem areas and the opportunity for **scientific and technical innovations** to resolve them, allowing to define a clearer picture on the features of the solution.	High
What should be given greater importance	School Doctor Interview	Appendix A.22, A.23	This investigation will explore the trade-off between two major factors affecting food choices, especially those of	High

when considering the diets of teenage school students - health or taste? [Debatable]			teenagers, and hence can provide insights into key reasons for the effectiveness of various diet and meal planners and programs. More importantly, it will help in synthesizing more relevant as well as helpful features of the eventual solution.	
How difficult is it for teenage school students to access equally healthy and palatable meals, especially from schools and meal planners? [Conceptual]		Appendix A.24	This investigation will delve into the crux of the problem by understanding the scale of challenges that schools and meal providers face in balancing the nutritional value as well as the delectability of the food they serve to teenage students. Therefore, the process of understanding the factors which cause these challenges will be made more seamless, and appropriate features for the solution can be accordingly synthesized.	High
What are the components of a balanced and healthy diet for teenage school students? [Factual]		Appendix A.25	This investigation will help identify the required aspects for a healthy and nutritious diet for teenage school students and serve as a benchmark for the required diet standards that a teenage school student must achieve on a daily basis. As a result, the scope of the problem can be further evaluated in order to understand the portion of the target audience consuming diets lacking nutritional value, also helping in inspiring an appropriate solution.	High
To what extent should schools be held solely accountable for the health and nutrition of their teenage students? [Debatable]	Interview with Fitness and Nutrition Coach	Appendix A.26, A.27	This investigation will examine the connection between the nutritional programs in schools to the root cause of the problem. Since teenage students consume a majority of weekly meals at their schools, this exploration will provide fruitful results in identifying the role of schools in influencing the **nutrition** and **well-being** of their teenage students. Thereby, a major possible source for the problem can be analyzed, with this synthesis also allowing to develop an effective solution later.	Medium
Analyse the relationship between the exercise and diet of teenage school students. [Conceptual]		Appendix A.28	This investigation will assist in understanding the components of a diet which is personalized to an individual and **adapt** to their individual requirements, by accounting for factors such as exercise and their effect on the diets of teenage school students. Throughout the exploration, keeping in mind the importance of this relationship will allow further study of factors fueling the problem, and will provide invaluable insights that can be employed when developing the solution.	Medium

By thoroughly investigating and analysing the above research questions, which were highly relevant to the previously determined problem states, I was able to employ an extensive range of primary and secondary sources in order to derive invaluable insights to understand the essence of the problem, the factors associated with it, as well as possible ways to act upon it, all through raw, qualified, and detailed, origins. **The evidences and summary for the above research can be found in Appendices A, B, and C**.

Strand (iii) Analyze a group of similar products that inspire a solution to the problem.

In order to examine the group of pre-existing products working in the defined problem area, I employed a SWOT (Strengths, Weaknesses, Opportunities, and Threats) Analysis, helping in generating invaluable insights on the characteristics and elements that my final solution for the problem needed to have and employ.

Product One: SuperCook - https://www.supercook.com/#/desktop

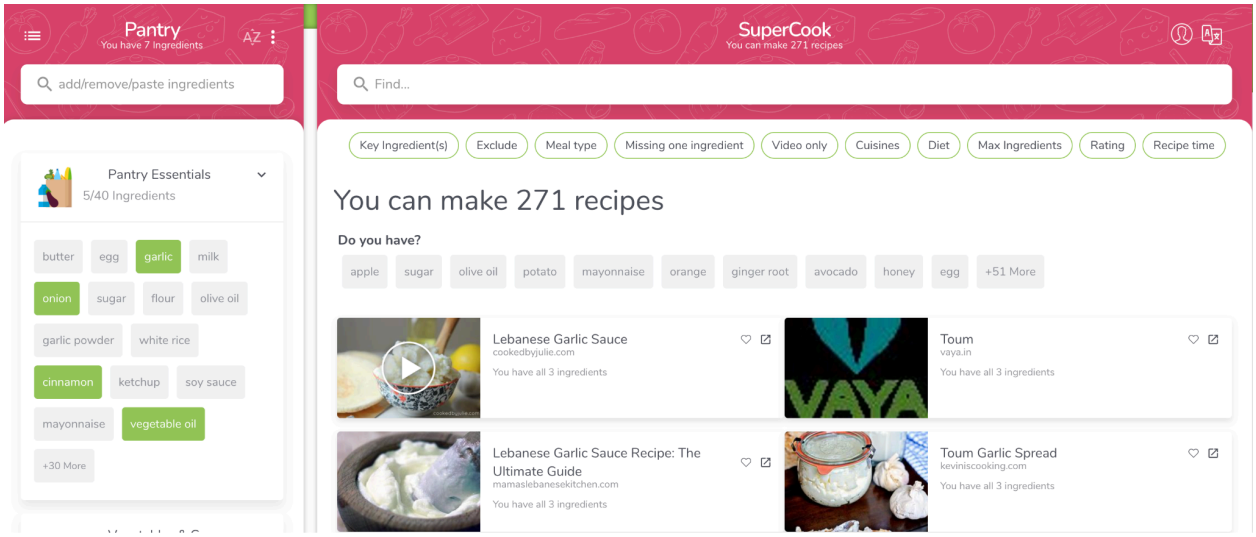


Fig: Supercook Website Images

SuperCook is a recipe search engine which allows users to input the ingredients they have readily available at home, following which it filters out options to suggest recipes and meals which can be made using those items, allowing for reduced food wastage, as well as a sense of personalisation.

Strengths	**Weaknesses**
• SuperCook is a free-to-use service involving no costs for users, which is mainly achieved by earning revenue through advertising revenue, ultimately allowing the tool to be highly accessible to the user group of teenage school students. • The website offers a sense of personalisation, rather than relying on the conventional one-size-fits-all approach, by allowing users to obtain a wide range of recipes and meals solely based on their unique requirements and scenarios, and thereby fostering a sense of **adaptiveness** to the various situations of teenage school students. • Users have options to filter the available recipes based on a range of parameters such as cuisines, diets, and meal type, offering a great amount of variety.	• The website tool has an extremely challenging and confusing user interface and user experience, causing numerous operational challenges as well as potential levels of frustration or irritation for the consumer group of teenage school students when they try to utilize the available features. • Furthermore, while SuperCook does offer a considerable quantity and quality of impactful features and applications, its complicated workflow makes it quite difficult for users to understand the true value and range of the features that the website has to offer, as well as the ways through which they can be utilized effectively. • The diet type filtering options which are available within the SuperCook product are rather generic and lack specificity, and don't offer significant value in terms of the customisation desired by the users.

Opportunities	Threats
• SuperCook is highly applicable in today's times because an increased portion of the customer base, and especially the target audience of teenage school students, is looking for quick and easy meals without significant effort, mainly due to rising professional as well as personal commitments. • There is also great scope for SuperCook to complement their current setup and offerings by adding features such as Artificial Intelligence-powered **personalized** recipe recommendations for teenage school students based on past preferences of ingredients in specific time frames and situations. • There is also ample possibility for the implementation of Generative Artificial Intelligence in order to dynamically generate custom recipes for teenage school students based on their ingredient availability, rather than depending on a pre-existing database, ensuring **personalisation** as well as innovativeness.	• These days, teenage school students are often relying on social media for diet tips and hacks, which can result in SuperCook potentially losing valuable customers. • Experiencing increased competition from numerous rising Artificial Intelligence-powered diet recommendation apps, who are able to provide an innovative value proposition. • Large Language Models (LLMs) such as ChatGPT by OpenAI and Gemini by Google Deepmind often give similar recommendations with a more seamless interface, and are already widely used by teenage school students for a variety of personal and professional tasks. • Together, these issues will all universally result in SuperCook losing the valuable customer base that it has tirelessly developed to other competitors who are able to provide similar features, but with an improved user interface and experience.

Product Two: Blue Apron - https://www.blueapron.com/

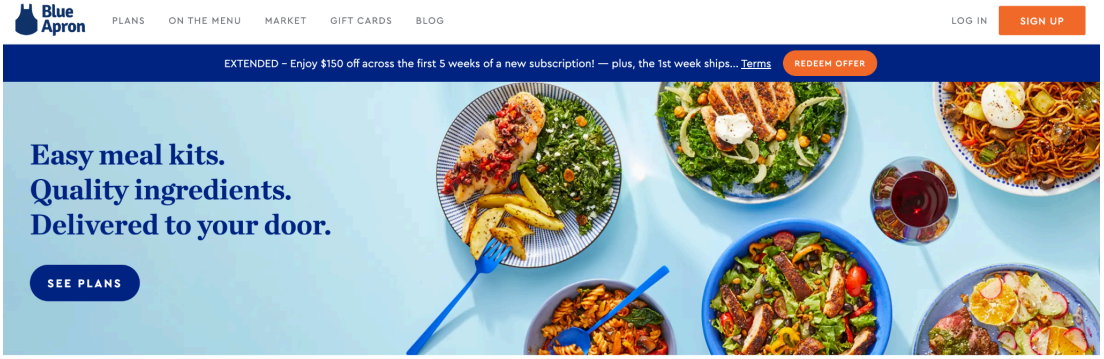


Fig: Blue Apron Website Images

Blue Apron is an online meal kit delivery service, through which subscribers receive fresh, already-measured ingredients along with an instructional recipe in order to cook fulfilling meals, specialising in catering a diverse range of meal options customized to the users.

Strengths	Weaknesses
• Blue Apron's offerings include a highly diverse range of simultaneously healthy and palatable food options that are available, which can be delivered directly to the doorsteps of customers, consequently contributing positively towards the company's sense of accessibility. • Blue Apron's overall concept and idea simplifies the process of cooking healthy and palatable meals and	• Due to the seamless delivery services that Blue Apron is able to provide, there is a notable downside visible, which is the fact that they often charge high prices for their products and services when compared to the ingredients at general stores. This makes their offerings rather inaccessible to many individuals,

makes the entire experience much more enjoyable as well as memorable, allowing them to sell emotions and memories along with products and services, to help in increasing customer satisfaction and subsequent retention. • Blue Apron provides unique features to customers such as allowing users to build their own personalised and customised diet and nutrition plans, while fostering an intuitive user experience.	especially for teeagers who are relatively financially dependent. • Furthermore,the provided meal plans eventually get repetitive and result in customers, especially teenagers, losing their sense of attraction towards the previously interesting recipes, and thereby derailing the overall impact of Blue Apron. • A lot of packaging and transportation goes behind such a model, potentially causing widespread environmental damage in the form of widespread carbon dioxide emissions.
Opportunities • They can further expand their business model as well as increase their revenue and subsequent profit by collaboration with grocery store chains in order to sell their meal kits in their spaces, allowing for increased visibility and reach. • Blue Apron can also integrate quick commerce models by taking advantage of newly efficient supply chain networks and dynamics through aspects like warehouses in order to provide faster service. • With the rise of eco-friendly and biodegradable packaging, they can work upon their environmental shortcomings in order to effectively provide their services while reducing their carbon emissions, allowing to increase their brand value as well as customer satisfaction and subsequent retention.	**Threats** • There is intense competition in the area of meal kit delivery with innovative business models, products, and services constantly arising, which can cause significant derailment for Blue Apron and its progression. • Quick commerce applications such as Instacart and Blinkit have revolutionized the delivery of necessities, being able to deliver a wide range of groceries and ingredients in fast durations, sometimes in less than 10 minutes, possibly resulting in the customer base switching from Blue Apron and utilizing such services. • With volatile as well as rapidly evolving supply chain networks, there is a possibility for a rapid rise in Blue Apron's operating costs, in terms of logistics and acquisition of raw materials, thereby consuming a significant portion of the revenue and reducing the overall profit.

Product Three: EatingWell - https://www.eatingwell.com/



Fig: EatingWell Website Images

EatingWell is an online website and magazine which assists in providing helpful information on healthy-eating, nutrition, and meal planning. Through blogs, articles, as well as numerous subpages, it offers nutritious and palatable recipes and diet plans for a variety of situations and requirements.

Strengths	**Weaknesses**
• EatingWell's selections provide a wide range of recipes that offer nutritious value along with a sense of variety, available in a plethora of categories, including meals for different times of the day, occasions, cooking setups and scenarios, as well as popular diet trends. • EatingWell's recipes and diet plans also help in catering to the diverse nutritional requirements and diets of its users for various health purposes, such as Mediterranean and Anti-Inflammatory Diets, along with specialized diets for patients with Diabetes. All of these diet types and planners have their individual purposes, being applicable for varying lifestyles of individuals as well as different aspects of one's health and well-being. • The website offers most, if not all, of its recipes and diet plans free of cost, making it highly accessible for teenage school students to implement for the betterment of their personal nutrition, **development**, and well-being.	• Despite significant advantages, EatingWell is still relatively aligned with the one-size-fits-all approach. While it does have a lot of options which can cater to the individual needs of the customer, they still can't satisfy their unique situations and requirements completely, being unable to provide the desired sense of **personalisation**. • Features a cumbersome user experience, which is highly complicated, featuring an excess amount of information and graphics in every page, and thereby lowering the quality of the overall website. • This graving issue can potentially lead to users, especially teenagers, who are unable to satisfy their necessities of seamless diet recommendations, looking for simpler, easier, and more effective alternatives.
Opportunities	**Threats**
• EatingWell can work on adding Artificial Intelligence elements along with their resultant diet and recipe plan customisation features, in order to capitalize on the upcoming trend of **personalized nutrition** and provide their consumers with a refreshing sense of **adaptiveness**, • EatingWell can also explore expanding their current model by building partnerships and integrations with other health and fitness apps and providers such as Apple Health, Peloton, and MyFitnessApp in order to provide more effective and informed recommendations to teenage school students, taking into account their fitness and activity in order to help in generating better nutritional insights. • There is also massive scope for EatingWell to integrate e-commerce into its existing business model, by exploring the selling of their own brand of kitchen appliances or cookbooks. Through this approach, a greater profit can be generated aside from the pre-existing ad-revenue, which can then be repurposed towards improving the user interface and experience.	• With teenagers spending more time on social media and relying on creators for diet tips and hacks that they relate to, the amount of EatingWell content consumed by teenage school students can decrease remarkably, resulting in issues such as lesser visibility and reach. • There is also immense competition in this domain of online solutions providing nutritious and palatable diets, recipes, and planners, with numerous health websites, magazines, and blogs all circulating similar content. This can end up causing severe concerns for EatingWell, with alternative solutions often having different improved features and experiences, taking a notable portion of their customer base away. • EatingWell's offerings mainly include diet and recipe planner recommendations featuring premium ingredients, meals, and health advice. However, with numerous countries experiencing rapid inflation rates today, large portions of the customer base will be much more reluctant to purchase the ingredients and machinery required by EatingWell, most probably resulting in teenagers not being able to access the offerings they have.

Strand (iv) Develop a design brief, which presents the analysis of relevant research.

After meticulously conducting relevant research through primary sources, including surveys, interviews, and workshops, along with secondary resources such as online articles and research papers, as well as a thorough analysis of numerous other existing products working within a similar problem, area I was able to narrow down the factors, aspects, and dynamics associated with the problem of nutritional solutions for teenagers lacking palatability and personalisation, as well as adaptiveness and accessibility, which consequently assisted in deriving invaluable insights on the components as well as qualities that my eventual solution needed to include, which are listed below.

My conclusions on the required elements of the solution, as a result of the primary and secondary research as well as analysis of existing products conducted, are:

1. Needs to be provided free-of-cost to the target user group of teenage school students in order to ensure accessibility and cost-effectiveness as well as universal visibility and reach, so that their financial dependence doesn't limit them from improving their nutrition and well-being.
2. Able to provide a diverse range of diet recommendations to increase variety and avoid repetition, and more importantly the loss of customer satisfaction.
3. Any diets or recipes need to be outlined with immense detail to provide the students with a seamless experience.
4. Must feature a simple User Experience which is self-explanatory and doesn't have any complicated or cumbersome aspects.
5. Diet and recipe recommendations need to be spontaneous and adaptive to different levels of physical activity as well as mental conditions, to ensure personalisation.
6. Needs to be developed from a qualified as well as exhaustive database, which predominantly consists of meals which balance both health and palatability.
7. Foster a sense of ethos and reliability, by implementing certified as well as dietician-backed nutritional frameworks as well as presenting diets and recipes in short video formats, with durations of 1-3 minutes.
8. Diverge from the one-size-fits-all approach and focus on customising diets and recipe plans for the individual requirements and scenarios of teenage school students.
9. Create an appealing environment which allows teenage school students to be deeply immersed, actively involved, as well as highly knowledgeable with regards to the planning and consumption of their diets.
10. Leverage Artificial Intelligence-based recipe recommendations and generations in order to ensure seamless and effective personalisation. However at the same time needs to make sure that it is not similar to general-purpose industry-common Large Language Models, which can be achieved using model fine-tuning as well as knowledge extension.
11. Be a completely online-based solution in order to avoid significant carbon emissions due to dynamics such as packaging, transportation, as well as high operating costs due to supply chain challenges.

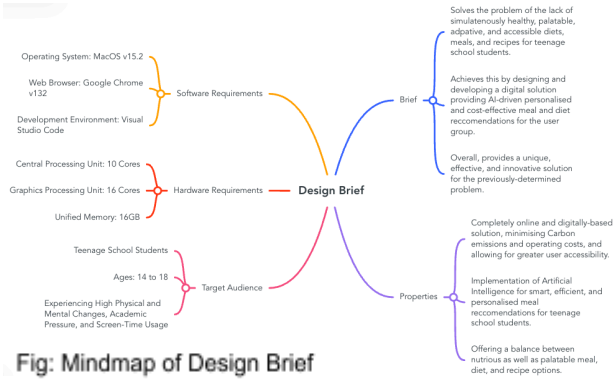


Fig: Mindmap of Design Brief

Furthermore, another interesting realisation that I had made, which I then identified as a necessary workflow as a part of the solution, this was that teenage school students consume their meals either at home, or outside, and the **flexibility** that they have in choosing the elements of their meal options vary significantly in both the different setups. This led to me to outline the fact that the digital solution needed to account for both these aspects in order to universally cater to the target audience. If they are looking for at-home meal recommendations, the solution should take into account the ingredients that are readily-available when making diet or recipe suggestions. However, if they are eating outside, either at school or a restaurant and they can't control the meal options available, the solution must calibrate the optimal meal recommendations

based on selections offered, all in order to ensure personalisation and adaptiveness. I culminated these findings into a visualised design brief (**Figure**), in order to serve as an appropriate guide while working on the development of the digital solution.

Criterion B - Developing Ideas

Strand (i) Develop a design specification, which outlines the success criteria for the design of a solution based on the data collected.

Criterion	**Specifications**	**Score**	
Aesthetics	The digital solution must feature a warm, geometric, sans-serif font style.	/2	/8
	Clear as well as informative graphics and images throughout the pages of the digital solution.	/2	
	The digital solution must use a light, calming, and balanced colour scheme.	/2	
	Overall minimal and sophisticated appearance of the final digital solution.	/2	
Cost	Monetary costs as a part of the developmental process should only be limited to purchasing necessary API keys, hosting and deployment services, and domain names.	/2	/8
	The total monetary cost incurred cannot exceed 20 United States Dollars.	/2	
	Time costs as a part of the developmental process should only be limited to ideating and researching the technical aspects required for the digital solution, programming the digital solution, and testing its performance.	/2	
	The total time cost incurred cannot exceed 20 hours.	/2	
Customer	The digital solution must cater to the predetermined demands of the determined target audience of teenage school students, within the age range of 14 to 18 years.	/2	/8
	The provided content as well as structure of the digital solution must offer a sense of credibility and reliability, by ensuring that it is certified, relatively short in duration, and highly-engaging.	/4	
	Offers a simple and highly effective User Interface, avoiding being cumbersome and complicated.	/2	
Functionality	The digital solution's tasks need to involve making effectual, seamless, informed, and cost-effective meal, recipe, and diet recommendations of palatable nutritional solutions to teenage school students.	/2	/8
	The recommendations and generations provided by the digital solution must be based on the personalised, spontaneous, and adaptive calibration of key data points, including available ingredients or meal options, intensity level of physical activity completed or expected, along with current mood, cuisine and dietary preferences.	/3	
	The program of the digital solution must be powered by an Artificial Intelligence-based workflow which assists in the dynamic recommendation and generation of meals and recipes for teenage school students.	/3	

Accessibility	The digital solution must be available to the user group of teenage school students free-of-cost, due to their financial dependence.	/2	/8
	The program of the digital solution needs to be implementable in versatile setups, including one's home, their school, or a restaurant, by having different features which are adaptive and personalized to the users' unique circumstances and requirements.	/3	
	The final digital solution has to be effectively accessed in numerous devices such as laptops, mobile phones, and tablets, as well as different browsers such as Google Chrome, Safari, Microsoft Edge, and Mozilla Firefox.	/3	
Safety	The digital solution must not access or store any Personally Identifiable Information (PII) or Health Data of the users, except their full name, email address, or phone number, only when required.	/2	/8
	The digital solution must not output inconsistent, inaccurate, or non-scientific information and advice to the target audience group. Any Artificial Intelligence-based systems cannot be implemented using general training data, in order to prevent hallucination.	/2	
	Any information or data that needs to be stored or transported will be done only using secure, reliable, and industry-acclaimed secure servers or protocols.	/2	
	Must ensure that any Large Language Model API employed will not implement the user's inputs for training processes.	/2	
Navigation	The final digital solution must have no less than 4 and no more than 6 main pages in order to ensure simultaneous elaboration and conciseness.	/4	/8
	Features an understandable as well as clear and organised standout navigation/menu bar.	/1	
	The digital solution must have straightforward and simple structure and navigation, allowing users to utilize every one of its aspects, while helping in avoiding confusion.	/3	

Taking the previously-determined design brief as a foundation, I then developed a success criteria (**Figure**), incorporating the highly-effective and widely-used ACCESSFM framework. This helps in quantitatively evaluating the effectiveness of numerous potential design ideas, also being useful when examining the success of the final digital solution. The features of the design ideas and solutions will be evaluated on each specification within the outlined list of criteria, resulting in the achievement of a cumulative score. They will then be cross-referenced with the grading table (**Figure**), which displays the resultant grades of each final score range, also allowing a qualitative evaluation, and a subsequent holistic understanding on the individual efficacy of the designs, in order to make more informed design selections. Furthermore, I then outlined a table featuring a diverse range of quantitative and qualitative testing methods (**Figure**), for the final outcome of the **developed** digital solution. Within this plan, I prioritized a sense of reliability as well as qualification and ensured to opt for appropriate sources and methods which reflected these features. Through these testing methods, my aim will be to evaluate the culminated digital solution, after designing product wireframes and implementing my technical skills, through an end-to-end lens, making sure I cover every single aspect of the success criteria I have developed, while obtaining invaluable, and more importantly bankable insights on the effectiveness of my digital solution.

Final Score	Grade
51-56	Exceptional
41-50	Satisfactory
33-40	Mediocre
25-32	Underwhelming
17-24	Weak
0-16	Ineffective

Fig: Mark to Grade Key

Criterion	Testing Method
Aesthetics	Acceptance Testing - Target Audience Survey
Cost	Accounting Sheets
Customer	Acceptance Testing - Target Audience Interviews
Functionality	Beta Testing - Peer Black Box Testing
Accessibility	Compatibility Testing - Different Devices and Browsers
Safety	Expert Evaluation - AI Engineer Appraisal
Navigation	Beta Testing - Peer White Box Testing

Fig: Testing Methods

Strand (ii) Present a range of feasible design ideas, which can correctly be interpreted by others.

Multiple design ideas for the digital solution, featuring a variety of formats, modalities, and features, were formulated, with each one being evaluated using a **target audience survey** along with a supporting examination against the success criteria. The culmination of this task is presented below. For each design idea involving wireframes, only the one for the home page has been displayed, **refer to Appendix D for the mockups of the rest of the involved pages.**

Design Idea One - Website

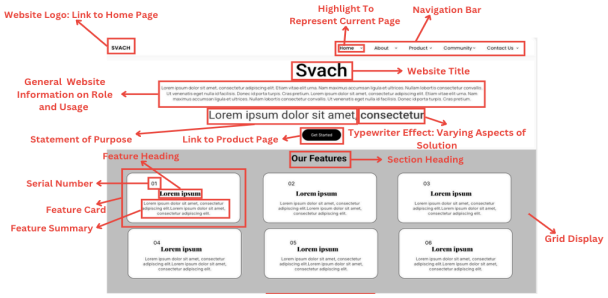


Fig: Home Page for Design 1

The first design idea created showcases detailed and annotated wireframes for a website, featuring five different pages. The Home page allows users to obtain general information on the website, which is called Svach, including its purpose, aspects, as well feature offerings. It also has a button linking the users directly to the product page, where the main implementation of the solution and its function occurs. This is followed by the About page, which gives the users a detailed account into the history and mission of the website, along with the problem it solves. It also gives an informational account into the website's main product and tool, which is Svach Servor 1.0, educating the users about its emergence and applications.

The main page is the Product page, which provides the users an opportunity to interact with Svach Servor 1.0, the website's focal function. In this page, users fill in the provided form in order to share their current preferences and circumstances, following which an Artificial Intelligence-based workflow will output personalised and relatable recipes, meal, and diet recommendations. It also has an option to switch from home mode to school mode accommodating the differences in both environments.

The website also includes a Community page, which engages its users with various forms of content and entertainment, in the form of short posts, opinion polls, and quizzes on nutrition, ultimately making the process of fulfilling one's nutritional requirements much more enjoyable and fulfilling. Finally, the website is completed with a Contact Page, allowing users to reach out for any concerns, questions, or suggestions they may have, featuring an efficient workflow which allows to obtain relevant data, including user information and contact details, as well as their central demands or requirements.

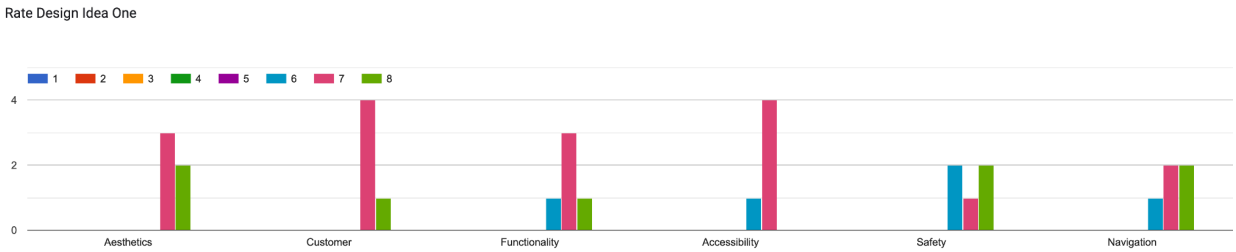


Fig: Survey Results For Design 1

Aspect	Summary of Evaluation	Score (/8)
Aesthetics	The digital solution features a warm font style, but a rather colour scheme, contributing to a minimal appearance. The graphics and images are all clear and informative.	6/8
Cost	The expected monetary cost is limited to necessary API keys, deployment services, and domain	6/8

	names. The expected time cost is limited to designing wireframes, researching for the product, and developing the digital solution.	
Customer	Caters to the identified problem area of teenage school students, aged 14 to 18. Credible and relatable content, using verified dietary frameworks, which are presented to the users. Highly simple and seamless User Interface.	7/8
Functionality	Powered by an Artificial Intelligence-based model to make effective meal, recipe, and diet recommendations to teenage school students, taking in a diverse range of data points.	6/8
Accessibility	Doesn't require to charge any payments for its usage, it is free-of-cost. Can be implemented in diverse scenarios, including home and school setups. Accessed in different types of devices and web browsers.	7/8
Safety	The components of the digital solution only access the name, email address, and contact information of the users, only when required. Any other information obtained is solely for the purpose of improving the product's functionality.	5/8
Navigation	The digital solution has between 4 to 6 pages and involves a standout navigation bar. Features straightforward and effective navigation, avoiding confusion.	7/8
Final Grade: Satisfactory		44/56

Design Idea Two - Mobile Application

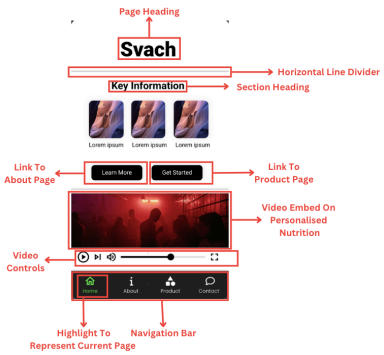


Fig: Home Page for Design 2

The second design idea formulated involves detailed and annotated wireframes for a mobile application, featuring four different screens. The first screen is the Home screen, featuring integral information on the purpose and characteristics of the digital solution, called Svach. It also provides multiple links in the form of buttons to other screens, permitting users to uncover more details about the application, as well as try out its product. It is completed with an embedded video, outlining its goal of providing the users with personalized nutrition, along with an introduction to some of its key techniques and methods.

The next one is the About screen, which provides the users with different concise accounts surrounding the digital solution and its purpose. This features sections on the history, mission, as well as core values of the application, presenting itself to the users as a credible and reliable solution. Next, it has the Product screen, which introduces the main functionality of the application, Svach Servor 1.0. Within this screen, users are first given a brief introduction into the capabilities and offerings of Svach Servor 1.0, following which they fill in a form with their dietary preferences and circumstances, which results in an Artificial Intelligence-based algorithm traversing through a database to provide optimal dietary recommendations to the users. Finally, the application ends with the Contact screen, initially involving a short description of how the screen may be of assistance, after which the users have the options to fill in a form, requiring their name and other contact information, as well as space for their main questions or points.

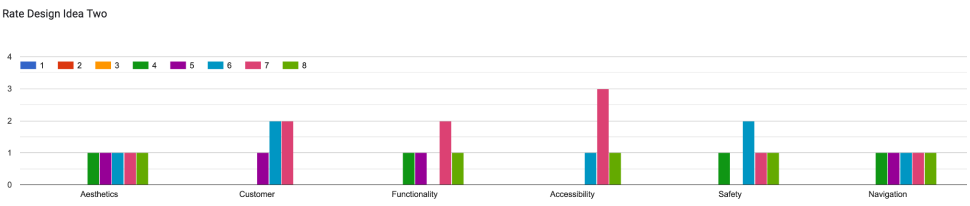


Fig: Survey Results For Design 2

Aspect	**Summary of Evaluation**	**Score (/8)**
Aesthetics	The digital solution features a warm font style, but a rather heavy colour scheme, contributing to a less minimal appearance. The graphics and images are all clear and informative.	5/8
Cost	The expected monetary cost is limited to necessary API keys, deployment services, and domain names. The expected time cost is limited to designing wireframes, researching for the product, and developing the digital solution.	6/8
Customer	Caters to the identified problem area of teenage school students, aged 14 to 18. Credible and relatable content, using verified dietary organisations, which are presented to the users. Relatively simple and seamless User Interface.	6/8
Functionality	Powered by a limited Artificial Intelligence-based model to make effective meal, recipe, and diet recommendations to teenage school students, taking in a diverse range of data points.	6/8
Accessibility	Doesn't require to charge any payments for its usage, it is free-of-cost. Accessed in a few different types of devices such as mobile phones and tablets.	6/8
Safety	The components of the digital solution access the email address and contact information of the users, only when required, but also access their full name, rather than just their first name. Any other information obtained is solely for the purpose of improving the product's functionality.	5/8
Navigation	The digital solution has 4 pages and involves a standout menu bar. Features straightforward and effective navigation, avoiding confusion.	6/8
Final Grade: Mediocre		40/56

Design Idea Three - Browser Extension

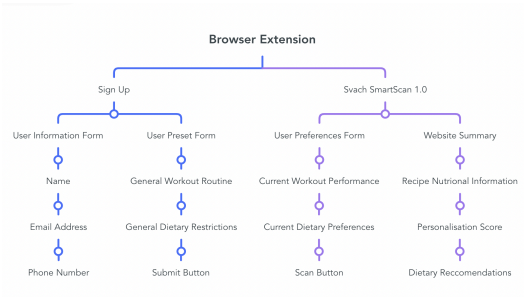


Fig: Sitemap for Design 3

The third and final design idea produced involves a detailed and exhaustive sitemap for a browser extension, taking an innovative approach by adding multiple pages to it. This design idea introduces an innovative tool known as Svach SmartScan, constituting the primary aspect of the digital solution. It features a simple workflow, consisting of only two pages which help it to execute its functionality, including the Sign Up page and the Svach SmartScan page.

The Sign Up page, involves two forms, both of which assist the browser extension by allowing to obtain relevant data to personalize the user experience, accommodating to their restrictions and requirements. The first form is a user information form, acquiring useful details on the user, including their name, email address, as well as phone number, allowing to personalise their dashboard, verify their identity, as well as relay any potential instructions. This is followed by a user preset form, which stores data on the non-negotiable preset aspects of the user's lifestyle, with regards to diet and exercise, also aiding in catering to a personalised experience.

After filling the forms in the first page, the user is then permanently led to the main product page, which is Svach SmartScan 1.0, a tool which conducts web scraping, extracting text data from various recipe websites by accessing the Document Object Model (DOM). Its functionality involves the users starting a new session, and then filling in a form which requests for their current dietary preferences and workout performance, allowing to personalise the results. This data is further appended to the initial preset data, and after the Scan

button is clicked, the extension will scrape the recipe website's text and cross-reference it with the presets and preferences. Finally, the extension will display key statistics and parameters describing the quality of the recipe website, in the context of the user's presets and preferences, including an overall appraisal on its nutritional and taste information, a personalisation score, and the ultimate recommendation regarding whether to use the recipe or not. This allows to instill a sense of personalisation and efficiency.

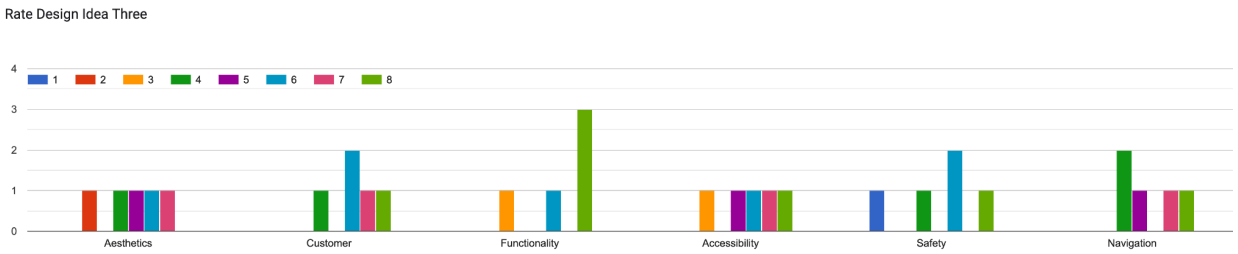


Fig: Survey Results For Design 3

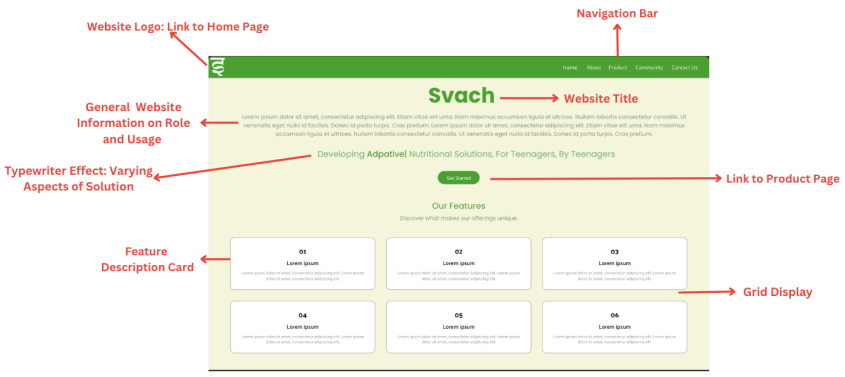
Aspect	**Summary of Evaluation**	**Score (/8)**
Aesthetics	The digital solution's overall workflow and modality limits the overall opportunity for graphics and images as a part of the browser extension. However, it can have a minimal and balanced colour scheme.	4/8
Cost	The expected monetary cost is limited to necessary API keys, deployment services, and domain names. The expected time cost is limited to designing wireframes, researching for the product, creating manifest files, and developing the digital solution.	6/8
Customer	Caters to the identified problem area of teenage school students, aged 14 to 18. Relatively short and highly-engaging content, through the forms and functionality availab;e. Relatively simple and seamless User Interface.	6/8
Functionality	Powered by a limited Artificial Intelligence-based model to make effective meal, recipe, and diet recommendations to teenage school students, taking in a diverse range of data points.	6/8
Accessibility	Doesn't require to charge any payments for its usage, it is free-of-cost. Can be implemented in diverse scenarios, including home and school setups. Accessed only in laptops as well as limited browsers.	5/8
Safety	The components of the digital solution access the email address and contact information of the users, only when required, but also access their full name, rather than just their first name. Any other information obtained is solely for the purpose of improving the product's functionality. Due to the fact that its functionality revolves around accessing the Document Object Model it visits, it may result in obtaining irrelevant and potentially private information or text.	5/8
Navigation	The digital solution has only 2 pages and thereby doesn't require a navigation bar. Features straightforward and effective navigation, avoiding confusion.	5/8
Final Grade: Mediocre		37/56

Strand (iii) Present the chosen design and outline the reason for its selection.

Criterion	Design Idea One	Design Idea Two	Design Idea Three
Aesthetics	6/8	5/8	4/8
Cost	6/8	6/8	6/8
Customer	7/8	6/8	6/8
Functionality	6/8	6/8	6/8
Accessibility	7/8	6/8	5/8
Safety	5/8	5/8	5/8
Navigation	7/8	6/8	5/8
Final Score:	44/56	40/56	37/8
Final Grade:	Satisfactory	Mediocre	Mediocre

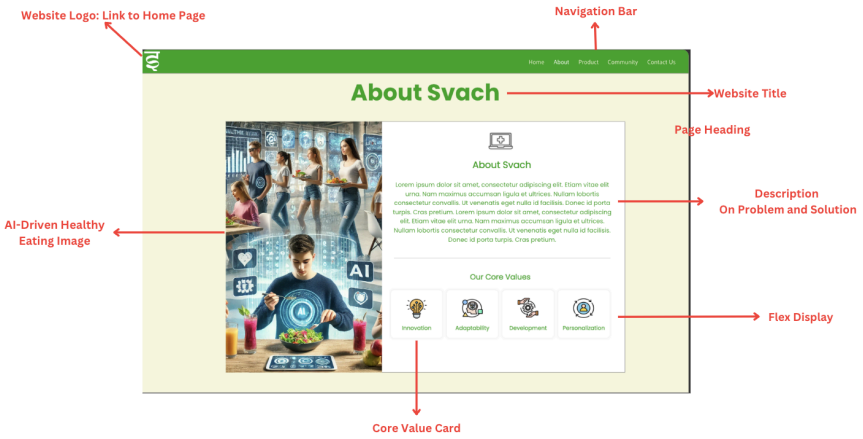
After a comprehensive and detailed evaluation of all the three formulated design ideas, using verified as well as unfiltered methods, especially through a target audience, I was able to conclude that Design Idea 1, the website, was the optimal foundation for the digital solution. It significantly outperformed Design Idea 2, the mobile application, and Design Idea 3, the chrome extension, especially in the aspects of navigation, accessibility, and aesthetics. However, I did identify various areas that possessed a scope for improvement, namely the aspects of safety, cost, and customer, all of which received similar lower-level evaluations.

Strand (iv) Develop accurate planning drawings/diagrams and outline requirements for the creation of the chosen solution.

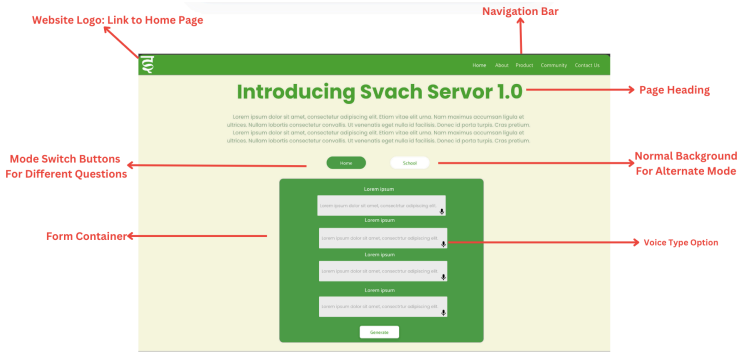


Home Page: The primary page of my digital solution, named Svach, will be the home page, serving as the base of all of its operations. The page will feature various static as well as dynamic elements, all to foster an immersive, yet simple, user interface and user experience. It will start off with a navigation bar, featuring links to all the five webpages in the website, with the current one being highlighted in a main colour. Furthermore, to the extreme left of the navigation bar, the website logo, which is the letter S, in the traditional Samarkan font, that also serves as a link to the home page from all the other webpages from the website. The home page features general elements such as a standout title for the name of the website and product, as well as a paragraph on what the digital solution is and how it works. Next, is the highlight aspect of the page, which is the statement of purpose, representing various qualities and characteristics that the solution aims to integrate, all using a typewriter effect. Furthermore, there is a button titled ‘Get Started’, which directly links the user

to the Product page, wherein they can interact with the main functionality of the digital solution, all through a simple and guided manner. Finally, the page will conclude with a list of various features that it integrates to solve the problem, using a grid display to appear appealing and engaging towards the user group of teenage school students.



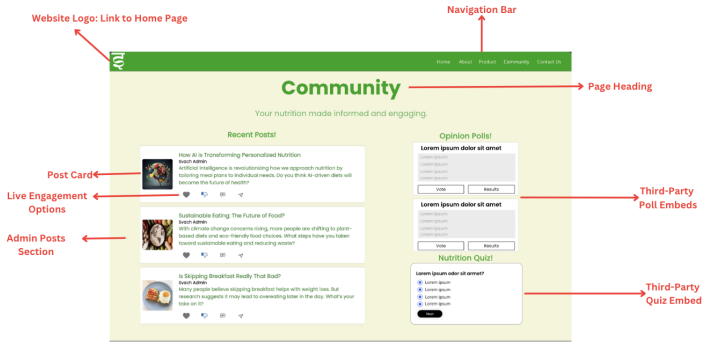
About Page: The second page of the website, is the about page, which allows to elaborately educate users about the purpose of the solution as well as the problem that it intends to solve, along with the main core values that it aligns with. It achieves this setup by using an overall grid display between an image of AI-based healthy eating for teenagers, a description of the solution and website, and the list of the core values which drive the website's functionality. An interesting fact about this webpage surrounds how it has an icon on the top of the description, of a cross symbol inscribed within a laptop, which signifies the use of technology in order to provide **healthy practices to shape the well-being of teenage school students**, aptly summarising the role and purpose of the website. Finally, a characteristic feature of this webpage is that it uses various aspects and elements to make it more engaging and aesthetically appealing to the target audience of teenage school students. This includes the use of bright colours and highly-realistic visuals from the hero AI-driven healthy nutrition image, along with the large images to represent the core values, along with the one-line texts for the same, all to ensure the website is simple to understand and traverse through.

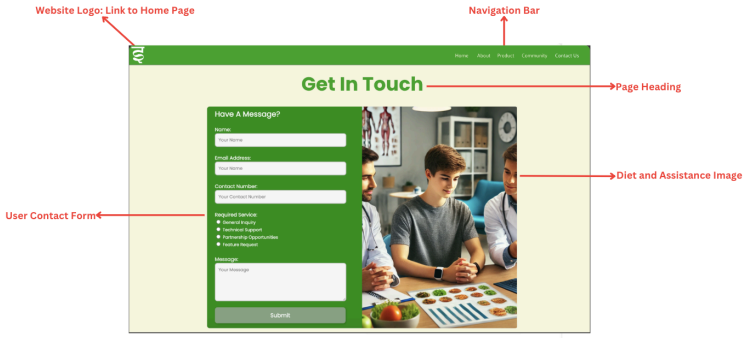


Product Page: Third, and most quintessential page of the website is the product page, which allows users to interact with Svach Servor 1.0, a novel, Artificial Intelligence-powered personalised and adaptive diet recommendation tool for teenage school students, ensuring cost-effectiveness, nutritional value, and palatability. It starts off with a standout heading text, introducing the product to the users, which is accompanied with a short and engaging paragraph, guiding the users on how to implement the solution in order to receive personalized meal and recipe plans or ideas. Next, the user is provided with two buttons options, which allow them to toggle between two different modes of utilizing the solution, which are home and school. Within the home mode, the user isn't faced with many restrictions for the type of meals they can cook, except for the available ingredients. In this mode, the user will input details on the available ingredients, along with other relevant information required for personalising the user's diets, such as workout routines, current mood and cravings, and cuisine preferences, following which an AI-driven system generates and provides appropriate and adaptive recipes. However, within the school mode, while a majority of the features stay the same as the home mode, the main change revolves around the fact that the student's are restricted to the meals that are provided by the school and need to work their options around that. So, instead of the available ingredients, the user will input all the different dishes available, along with data such as mood, cravings, workout routines, and cuisine, which will then be synthesized by the AI-based system in order to recommend the user with a meal plan of all the available dishes that matches their preferences. The school mode can also be applicable for restaurants, and other places where one may have to eat outside the comfort of their home, with the total options being limited. The use of this AI-driven system will ensure that the digital solution will be more cost-effective for the users, because minimal time as well as research and development costs will be incurred for its development,

allowing it to be more cost-effective for the users. Also, to ensure that the content presented to the users isn't cumbersome, the meal suggestions to the users will be in the form of short videos up to five minutes in duration, that will be overlaid upon the screen in order to ensure that the solution is engaging. Furthermore, the options for the users to toggle between both the home and school modes, will make sure that the digital solution can be implemented in a variety of scenarios and environments, always being by the side of the user. Together, both of these elements allow this webpage, and the overall digital solution of the website to be highly accessible for the users.

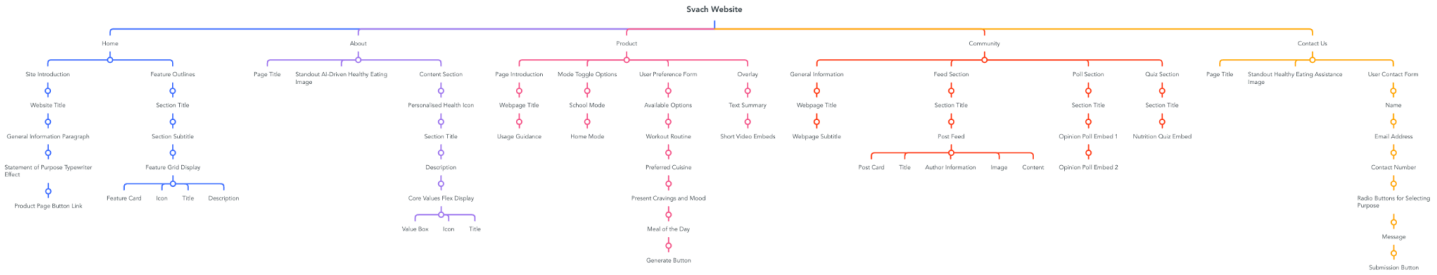
Community Page: The fourth and penultimate page of the website is the community page. Within this page, the user can access and utilize several interactive elements, including a post feed, opinion polls, and a weekly nutrition quiz, making the webpage and overall appealing much more appealing to them, while also serving as a source of positive and constructive entertainment for them. Most importantly, it makes the whole process of nutrition engaging and interesting for the target audience of teenage school students. Furthermore, it allows to work upon the previously determined solution area of fostering a greater sense of teenagers being involved in their diet, nutrition, and **development,** as well as informed about the various scientific elements and dynamics associated with the consumption of their meals. The post feed will be regularly updated by the website administrator, featuring various informative suggestions and tips for teenagers to improve their nutrition and overall **well-being**, at the same time all being relatively short in terms of word count in order to guarantee the user with an engaging and seamless experience. The opinion polls will ask users for their thoughts on some debatable and popular topics within the realm of teenage nutrition and well-being, along with a display of live global results after answering. The nutrition quiz will test the user group of teenage school students on their knowledge of various factors influencing their nutrition, physical health, mental health, and overall well-being, providing the target audience with a chance to **gamify** the process of **developing** these imperative characteristics. Both the element of the opinion polls and the nutrition quiz will be embedded onto the website from expert, third-party options.





Contact Us Page: The fifth and final page of the website is the contact us page, giving an opportunity for the users to reach out to the administrators and creators of the digital solutions in order to clarify concerns, relay vital information that they would like to bring to light, or share details on potential bugs or inaccuracies. On the right of the webpage, there is a standout image of professionals assisting a teenager with his diet and nutrition, representing the fact that the purpose of the webpage is to provide the target audience of teenage school students with state-of-the-art expert advice and assistance with any queries or concerns that they would like to raise. The form takes the inputs of integral and necessary user data, such as their name, email address, contact number, purpose of query, as well as their main message. This is then followed by a submit button to complete the form, leading the user to an alternate overlay thanking the user for their submission and redirecting them back to the home page, thereby allowing the website to complete its full cycle. This system also allows the website to have an in-built feedback loop during its initial stages, obtaining relevant and highly effective feedback in order to improve the product, allowing to continuously improve the product to meet the demands of the target audience of teenage school students.

Sitemap:



Color Scheme:

18A414	F5F5DC	62B85F	058E00	82A481	FFFFFF	D3D3D3
Pigment green	Beige	Mantis	India green	Cambridge blue	White	Timberwolf

BoM Table:

S. No	Item	Cost (USD)
1	Visual Studio Code	-
2	Frontend Hypertext Markup Language Framework	-
3	Frontend Cascading Style Sheets Framework	-
4	Frontend Javascript Framework	-
5	Backend Python Flask Server	-
6	Jupyter Notebook	-
7	Youtube Recipe Playlists Datasets	-
8	Pandas, Numpy, OS Libraries	-
9	Youtube Data Developer API	-
10	OpenAI Developer API	10
Total Cost:		**$10**

Success Criteria Evaluation:

The above final wireframes were created after formulating the initial mockups of the website and obtaining relevant as well as helpful feedback from the target audience of teenage school students. I then evaluated them based on the overarching themes and expectations of the user group which I was able to derive from the insights and appraisals that I had obtained from earlier primary research and user interactions. Due to the fact that a few of the specifications of the success criteria could only have been evaluated once the final version of the digital solution was completed and tested using reliable testing methods, it was difficult to guarantee a full evaluation for those metrics. That is why, I was only able to partially evaluate those few cases to the best of my knowledge from the past appraisals and insights that I had obtained from experts and members of the target user group, setting the scope for a more effective evaluation following the **development** of the digital solution.

Criterion	Specifications	Score	
Aesthetics	The digital solution must feature a warm, geometric, sans-serif font style.	2/2	8/8
	Clear as well as informative graphics and images throughout the pages of the digital solution.	2/2	
	The digital solution must use a light, calming, and balanced colour scheme.	2/2	
	Overall minimal and sophisticated appearance of the final digital solution.	2/2	
Cost	Monetary costs as a part of the developmental process should only be limited to purchasing necessary API keys, hosting and deployment services, and domain names.	2/2	7/8
	The total monetary cost incurred cannot exceed 20 United States Dollars.	2/2	
	Time costs as a part of the developmental process should only be limited to ideating and researching the technical aspects required for the digital solution, programming the digital solution, and testing its performance.	2/2	
	The total time cost incurred cannot exceed 20 hours.	1/2	
Customer	The digital solution must cater to the predetermined demands of the determined target audience of teenage school students, within the age range of 14 to 18 years.	2/2	6/8
	The provided content as well as structure of the digital solution must offer a sense of credibility and reliability, by ensuring that it is certified, relatively short in duration, and highly-engaging.	2/4	
	Offers a simple and highly effective User Interface, avoiding being cumbersome and complicated.	2/2	
Functionality	The digital solution's tasks need to involve making effectual, seamless, informed, and cost-effective meal, recipe, and diet recommendations of palatable nutritional solutions to teenage school students.	1/2	7/8
	The recommendations and generations provided by the digital solution must be based on the personalised, spontaneous, and adaptive calibration of key data points, including available ingredients or meal options, intensity level of physical activity completed or expected, along with current mood, cuisine and dietary preferences.	3/3	
	The program of the digital solution must be powered by an Artificial Intelligence-based workflow which assists in the dynamic recommendation and generation of meals and recipes for teenage school students.	3/3	
Accessibility	The digital solution must be available to the user group of teenage school students free-of-cost, due to their financial dependence.	2/2	6/8
	The program of the digital solution needs to be implementable in versatile setups, including one's home, their school, or a restaurant, by having different features which are adaptive and personalized to the users' unique circumstances and requirements.	2/3	
	The final digital solution has to be effectively accessed in numerous devices such as laptops, mobile phones, and tablets, as well as different browsers such as Google Chrome, Safari, Microsoft Edge, and Mozilla Firefox.	2/3	
Safety	The digital solution must not access or store any Personally Identifiable Information (PII) or Health Data of the users, except their full name, email address, or phone number, only when required.	3/3	6/8

	The digital solution must not output inconsistent, inaccurate, or non-scientific information and advice to the target audience group. Any Artificial Intelligence-based systems cannot be implemented using general training data, in order to prevent hallucination.	1/2	
	Any information or data that needs to be stored or transported will be done only using secure, reliable, and industry-acclaimed secure servers or protocols.	1/2	
	Must ensure that any Large Language Model API employed will not implement the user's inputs for training processes.	1/1	
Navigation	The final digital solution must have no less than 4 and no more than 6 main pages in order to ensure simultaneous elaboration and conciseness.	4/4	8/8
	Features an understandable as well as clear and organised standout navigation/menu bar.	1/1	
	The digital solution must have straightforward and simple structure and navigation, allowing users to utilize every one of its aspects, while helping in avoiding confusion.	3/3	
Final Grade: Satisfactory		48/56	

Criterion C - Creating the Solution

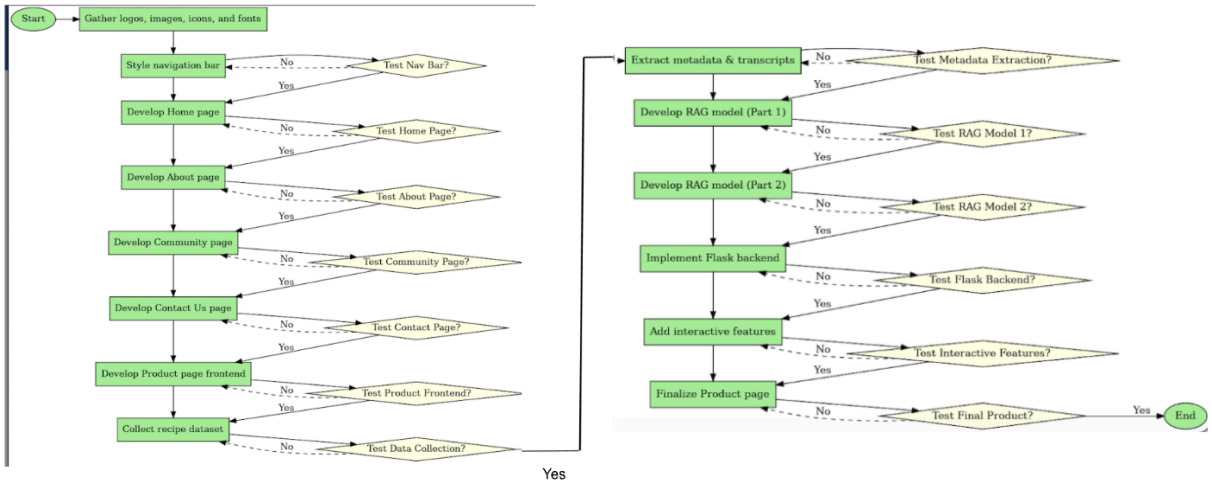
Strand (i) Construct a logical plan, which describes the efficient use of time and resources, sufficient for peers to be able to follow to create the solution.

Task	Justification	Resources	Duration (Hours)
Gather and create required logos, images and icons, and choose an appropriate font family.	Will ensure that the website is aesthetically pleasing as well as visually pleasing and engaging for the target audience of teenage school students. Will allow to make progress on the **Aesthetics** aspect of the success criteria.	Google Chrome, Google Fonts, Google Docs	0.5
Create and style the navigation bar of the website.	A stylish and moreover, functional navigation bar of the website will allow the user a sense of seamlessness with respect to traversing through the multiple features and pages of the websites. Will allow to make progress on the **Navigation** aspect of the success criteria.	Visual Studio Code, HTML, CSS, W3 Schools, Google Chrome	0.5
Develop the full functionality of the Home page.	Program and integrate the various formulated designs in order to present the intended features and ideas for the home webpage to the user group.	Visual Studio Code, HTML, CSS, JS, W3 Schools, Google Chrome	1
Develop the full functionality of the About page.	Program and integrate the various formulated designs in order to present the intended features and ideas for the about webpage to the user group.	Visual Studio Code, HTML, CSS, W3 Schools, Google Chrome	0.5

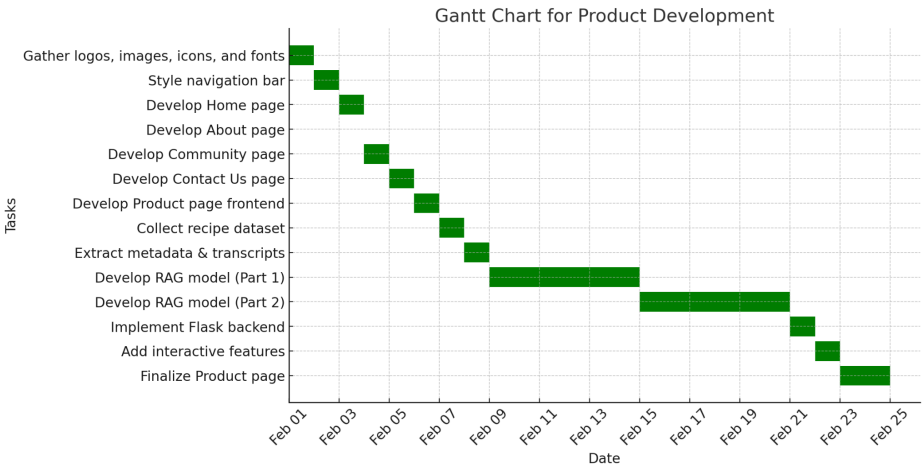
Develop the full functionality of the Community page.	Program and integrate the various formulated designs in order to present the intended features and ideas for the community webpage to the user group.	Visual Studio Code, HTML, CSS, W3 Schools, Google Chrome	1
Develop the full functionality of the Contact Us page.	Program and integrate the various formulated designs in order to present the intended features and ideas for the contact us webpage to the user group.	Visual Studio Code, HTML, CSS, W3 Schools, Google Chrome	1
Develop the Frontend of the Product page.	Program and integrate the various formulated designs in order to present the initial outline for the product webpage to the user group.	Visual Studio Code, HTML, CSS, W3 Schools, Google Chrome	1
Build recipe dataset by collecting video recipes.	Compile and synthesize video playlists and recipes from platforms such as Youtube and Vimeo, so that a Retrieval-Augmented Generation-based Large Language Model can effectively be trained and refined with numerous data points and in-depth knowledge, to provide recipe recommendations.	Youtube, Vimeo, Google Chrome, W3 Schools	1
Extract metadata and transcripts of the recipe video dataset.	Allows to seamlessly access reliable, helpful, quantitative, as well as informative data on the recipe dataset in order to train the Retrieval-Augmented Generation-based Large Language Model, in an easy and effective manner.	Youtube Data API, Visual Studio Code, MacOS Finder, Google Chrome	0.5
Develop Retrieval-Augmented Generation-based Large Language Model for the Backend of the Product page.	A Retrieval-Augmented Generation-based Large Language Model will allow to expand the knowledge of a traditional Large Language Model by providing additional verified data points and information, which will allow to make scientifically-backed as well as reliable recipe and meal recommendations to the user group of teenage school students. Will allow to make progress on the **Functionality** aspect of the success criteria.	Visual Studio Code, OpenAI API, Youtube, W3 Schools, Python, Jupyter Notebook, Pandas, Numpy, OS, Google Chrome	3
Implement Flask Server to receive user inputs from the Frontend of the Product page.	In order to transport the data from the frontend user input form, run it through the Retrieval-Augmented Generation-based Large Language Model, and revert back by displaying the meal and recipe recommendations to the user, a Flask server will be required to connect the frontend to the backend.	Visual Studio Code, Flask, Python, Jupyter Notebook, OpenAI API, JS, W3 Schools, Google Chrome	1.5
Integrate interactive features to the Product page.	Program and integrate the various intended interactive, dynamic, and high-technical features and ideas for the home webpage, in order to	Visual Studio Code, HTML, JS, W3 Schools, Google	0.5

	ensure a sense of seamlessness, implementability, as well as attractiveness for the user group of teenage school students. Will allow to make progress on the **Customer** aspect of the success criteria.	Chrome	
Complete the full functionality of the Product page.	Synthesize, integrate, and troubleshoot the various previously-developed features, components, and attributes for the product webpage, guaranteeing that it functions as intended and all the elements involved collaborate to provide a simple and effective interface and experience for the user group of teenage school students.	Visual Studio Code, HTML, CSS, JS, Flask, Python, Jupyter Notebook, OpenAI API, W3 Schools, Google Chrome	2
Total Duration:			**14 Hours**

Flowchart:



Gantt Chart:



Strand (ii) Construct a logical plan, which describes the efficient use of time and resources, sufficient for peers to be able to follow to create the solution.

Step 1: Frontend Content Development Using Hypertext Markup Language and Cascading Style Sheets

In order to effectively present the static and semantic content as well as elements, I utilized the programming languages of Hypertext Markup Language (HTML) and Cascading Style Sheets (CSS), with the programming environment of Visual Studio (VS) Code. Throughout the pages I implemented numerous standard HTML tags such as ** **, **<h1>**, **<p>**, **
, **<form>, and **<button>**, as well as CSS properties like **text-align**, **background-color**, **color**, **margin**, **width**, **height**, **font-size**, and **font-family**. One of the stronger implementations of HTML which I used throughout the website was for the navigation bar, wherein I utilize a series of tags such as ** **, **<ul>**, and **<a>**. I used the division or tag to create a separate container for the navigation bar, allowing me to implement specific and particular CSS styling for the navigation bar, thereby promoting a sense of efficiency. I used the unordered list or <ul> tag to create a list of all the five pages of the website, and inside each of the 5 <li> or list tags, I placed an <a> or anchor tag linking the user to each of the five pages. Next, I styled the same HTML content for the navigation bar using CSS, I included elements such as **list-style-type**, **display**, **text-decoration**, and **position** in order to ensure that the list and hyperlink setup would look aesthetic enough to match the final expected outcome of a navigation bar standard. Next, another implementation wherein I was able to heavily implement technical skills to **develop** a complex outcome was for the About Page, in which I implemented a container setup for three main elements, a hero image, a text description, and a side-by-side display of core value icons. This was mainly achieved using ** **, **<img>**, **<p>**, and **<section>** HTML tags, along with a **grid display** for the container, **flex display** for the core value boxes, as well as **padding**, **box-shadow**, and **border-radius** for styling the boxes, as my CSS properties, for the successful completion of the page.

Navigation Bar

```html
<div class="navbar">
  <ul>
    <li><a href="home.html" class="logo"><img src="logo.png" width="50px", height="70px"></a></li>
    <li class="active"><a href="home.html">Home</a></li>
    <li class="inactive"><a href="about.html">About</a></li>
    <li class="inactive"><a href="prod.html">Product</a></li>
    <li class="inactive"><a href="comm.html">Community</a></li>
    <li class="inactive"><a href="contact.html">Contact</a></li>
  </ul>
</div>
```

```css
.navbar ul {
    list-style-type: none;
    display: flex;
    align-items: center;
    padding: 0;
    margin: 0;
    width: 100%;
}
.navbar li {
  display: inline;
}
.navbar li a {
  text-decoration: none;
  font-size: 20px;
  padding: 10px 10px;
}
```

```css
.navbar {
  background-color: #18a414;
  overflow: hidden;
  top: 0;
  left: 0;
  right: 0;
  text-align: center;
  position: static;
}
.navbar ul li {
  display: inline-block;
}
.navbar ul li:first-child {
  margin-right: auto;
}
```

About Container

```html
<div class="container">
  <div class="main-image">
    <img src="aboutcoll.webp" alt="Main Image">
  </div>
  <div class="content-section">
    <div class="about-section">
      <img src="laptop.png" width="64px" heigh
      <h2>About Svach</h2>
      <p style="text-align: justify;">At Svach
    </div>
    <hr>
    <section class="core-values">
```

```html
<div class="value-box">
    <img src="personal.png" alt="Icon">
    <span>Personalization</span>
</div>
```

```css
.container {
    display: grid;
    align-items: stretch;
    grid-template-columns: 1fr 1fr;
    height: 100vh;
    max-width: 1000px;
    margin: auto;
}
```

```css
.value-box {
    width: 100px;
    height: 90px;
    display: flex;
    flex-direction: column;
    align-items: center;
    padding: 15px;
    background: white;
    border-radius: 8px;
    box-shadow: 0 0 5px rgba(0, 0, 0, 0.1);
}
```

Step 2: Frontend Interactivity Development Using Javascript

Even though a majority of the webpages relied mainly on static components to fulfill their respective functionalities, they did require a sense of interactivity in a few particular cases, and even though they were not highly complex in nature, they still helped in ensuing that the final outcome of the features were indeed aesthetically pleasing and purposefully successful in achieving the set plans. For those particular instances, the programming language of Javascript (JS) served as effective in fulfilling the involved requirements. The first case in which JS was the heart of the implementation was the **Typewriter Effect** in the Home page of the website. For this task, I implemented a function known as **typeEffect()**, which would use a preset list of various words, stored in the variable **texts**, to present using the desired typewriter effect, taking a word, represented by the variable **count**, from the list and then slicing it into its individual letters, and **appending** each one in an orderly fashion to produce a visual as if the word is being typed out. This was achieved by using the property of the **Document Object Model** and **getElementById**, in order to alter the content of the HTML webpage, through the JS. This is then followed by increasing the value of count by 1 in order to type in the next word, and then continuing a similar process until all the words are completed, after which the process will loop again, **iterating** infinitely. The second implementation, which wasn't highly complex and technical, but still required to give the user group of teenage school students an **appealing sense of**

engagement through **social media-esque interaction** buttons, were the options for user engagement. One such function that I had developed was known as **toggleLike()**, which accesses a link in the HTML file, placed inside a heart image in order to allow the user to **like** a post, used the **querySelector** property to extract the content image, checks whether the link has been clicked, and then changes the source or **src** of the image, to **liked.png** to display a red heart image, signifying that the user has liked the post, and to **like.png** To display a grey heart image, signifying that the user has not liked the post.

Typewriter Effect

```
function typeEffect() {
    if (count === texts.length) {
        count = 0;
    }

    currentText = texts[count];
    letter = currentText.slice(0, ++index);

    document.getElementById("typewriter-text").textContent = letter;
```

```
    if (letter.length === currentText.length) {
        count++;
        index = 0;
        setTimeout(typeEffect, 750);
    } else {
        setTimeout(typeEffect, 100);
    }
}
```

Post Engagement

```
function toggleLike(anchor){
    let img = anchor.querySelector("img");
    if (img.src.includes("like.png")) {
        img.src = "liked.png";
    } else {
        img.src = "like.png";
    }
}
```

Step 3: API-Driven Large Language Model Development and Prompt Engineering With Python

This step was one of the most important ones, for ensuring the successful **Functionality** of the digital solution, and especially its product **Svach Servor 1.0**, for it was the main implementation of the desired Artificial Intelligence workflow. I wanted to use a Large Language Model (LLM) Application Program Interface (API) which was fast and concise to provide the users with real-time recommendations, but also highly-accurate, deep, and knowledgeable, ultimately leading me to select the OpenAI API, and particularly the GPT 4o Mini model for this task. These recommendations needed to be based on the series of inputs provided by the user from the text areas in the product page. I acquired my own personal **API Key** from the OpenAI Developer Dashboard and initialized a **client** for the model using the API key. Next, I implemented a setup through which the model would generate a **response** to the user's **prompt**, which was mainly achieved through the **Chat Completions** workflow. This gave multiple messages to the selected model, including the **role**, which constituted to the individual requiring the model, or the user in this case, as well as the **content**, which essentially meant the prompt provided by the user. For this prompt inputted by the user, I developed a custom procedure to prepare the user's raw prompt from the HTML page, to a format suitable enough for the LLM to obtain enough information and instructions to make an accurate recommendation, by implementing **prompt engineering**. I stored the complete prompt in a python **f-string**, allowing for simplicity, as well as for the seamless integration of variables. The process involved sharing the previously synthesized dataset of meals to the model, mentioning that they are the available ones to recommend from, ensuring that the response is indeed accurate and valid. Following which, I gave explicit instructions to the model to use only the meal dataset to recommend the recipes, emphasizing on the point by capitalizing the word **ONLY**, as well as prompting it to provide the user with in-depth and simple explanations of the meal recommendations, along with suggestions to the users to further personalise their meals to their individual requirements, all helping to make the digital solution more **adaptive** and **efficient**.

OpenAI API Implementation

```
client = OpenAI(
    api_key = OPENAI_API_KEY
)
```

```
response = client.chat.completions.create(
    model = "gpt-4o-mini",
    messages = [
        {
            "role": "user",
            "content": prompt_prep(user_prompt, user_share)
        }
    ],
    max_tokens = 1200
)
```

LLM Prompt Engineering

```
    prompt = f"""
I want you to act as a meal reccomendation agent. We have the following meals:
{combined_text}

Based on the user prompt and the meals listed above, please provide three meals as reccomendations. Use ONLY the meal dataset to recommend meals. Do
Only respond based on the provided dataset. Do not generate unrelated recommendations.
If the user's prompt and unique requirements don't match the meal reccomendation completely, provide suggestions on how to achieve their needs, espe
Explicitly talk about how the meal can satisfy all of their prompt's needs and requirements.

Here is the user prompt {user_prompt}
"""

    return prompt
```

Step 4: Retrieval-Augmented Generation Using OpenAI API and Python

This step was the most important as well as highly-technical one out of all of them as it involved me venturing into a new area of Artificial Intelligence Development - Retrieval Augmented Generation (RAG). RAG helps in **extending** the knowledge of traditional Large Language Models (LLM), allowing it to progress from **generic training data** to **accurate** and **industry-specific information** from external sources and databases. I started off by using the Open AI API to create **vector embeddings** for the transcript of every recipe video in the dataset called **meals_data**, conducting this process by generating vectors in **512** dimensions to ensure a sense of depth and accuracy. After acquiring the embeddings for each meal in the dataset, I created a new column in the existing meals_data

dataframe, called **embdedding_vector**, wherein I stored the embeddings for each row of the different meals. I then initialized a function called **prompt_prep()**, to help **preprocess** the user prompt to find the **optimal** meal recommendation as well as to obtain the relevant data that would be used for prompting the Large Language Model. This function took in two parameters, **user_prompt** and **user_share**, both of which were the same string extracted from their responses in the HTML, with both being used for different purposes. The local variable user_share was used in the process of generating the vector embedding for their input, similar to how it was done for the meals dataset, using the model Text Embedding 3 Small, while the variable user_prompt was used for the final prompt engineering of the LLM. After obtaining the vector embeddings for the prompt, I created a variable called **user_share_vector** and used the **Numpy** library to convert it into an array, which I also did for the meals dataset, when I created a variable called **meals_vectors**. The next part involved using the evaluation metrics from the **Scikit-Learn** library, specifically **cosine_similarity**, which is used to find how closely two vector representations of the text **align**, by taking their cosine. I implemented the cosine_similarity function to find the alignment between the vectors for all the meals in the dataset with the user input's vector, following which, I created a new column in the dataset called **similarity_measure**, having a **float** value between 0 and 1 to represent how much each meal aligns with the user's input, with the higher number being the optimal one. I then sorted the dataframe based on the similarity_measure column, in a **descending order**, and then obtained the top three meals using the **head()** procedure, in a variable called **top_3_items**. Finally, I joined the names of the top three meals into a variable called **combined_text**, which I used during the prompt engineering along with initial **user_prompt** variable, indicating the model to present the meal options in combined_text based on the user's preferences in user_prompt, making it personalised, adaptive, and specific, also refining the LLM to ensure it **doesn't hallucinate**, or make inaccurate outputs, ensuring **safety** and **effectiveness.** This will be applicable for the product page, wherein all the inputs provided by the users in the form of the HTML will be synthesized by the backend, run through the RAG-based LLM to output a relevant and effective recommendation to the users.

Embedding Generation

```
response = client.embeddings.create(
    model="text-embedding-3-small",
    input = meals_data.item.tolist(),
    dimensions = 512
)

vector_embeddings = [d.embedding for d in response.data]
meals_data['embedding_vector'] = vector_embeddings
meals_vectors = [np.array(vec) for vec in meals_data.embedding_vector.values]
```

```
def prompt_prep(user_prompt, user_share):
    response = client.embeddings.create(
        model="text-embedding-3-small",
        input = [user_share],
        dimensions = 512
    )
    user_share_vector = [d.embedding for d in response.data]

    user_share_vector = np.array(user_share_vector)
```

Meal Selection

```
similarity_matrix = cosine_similarity(meals_vectors, user_share_vector)
similarity_values = [v[0] for v in similarity_matrix]

meals_data['similarity_measure'] = similarity_values
meals_data.sort_values(by = "similarity_measure", ascending=False, inplace=True)

top_3_items = meals_data.head(3)

combined_text = '\n\n'.join(top_3_items['item'])
```

Step 5: Backend Integration By Implementing Flask Server For User Input Handling Through Python

After developing my HTML, CSS, and JS fronted for the Product Page as well as the RAG-based LLM for the backend of the product page, I required a mechanism to transport and share data across both the frameworks, to ensure that the product would perform exactly as intended. In the fronted, this involved **extracting** the users' inputs from the HTML page using the **getElementByID** mechanism, formatting them into appropriate **f-strings** to obtain the **userInput** and **userShare** variables, then sending these inputs to the Flask server using a **Fetch API Request** and a **POST** method. Then in the backend, I set up a **generate** route to handle the POST requests, and using **request.json** to extract the JSON payload sent from the HTML and JS frontend for the arguments of **prompt** and **sharePrompt**, and storing them is variables called **user_prompt** and **user_share**, respectively, in order to input them into the RAG preprocessing and prompting function. Then after passing the prompts through the RAG-based LLM, I obtained data for the **Video ID** of each of the top three recipe videos identified by the similarity function. After which, I returned the AI-generated text as the response of the **OpenAI API call**, along with the three links, back to the user in a **JSON** form. Then, the frontend Javascript receives the **response** from the backend, and obtains all the required variables. It will then embed the three video IDs of the personalized recipe videos in the **Practice** section of the Product Page's home mode, along with the AI-generated response from the RAG-based LLM in the **Plan** section of the Product Page's home mode, thereby displaying the final recommendation outcome to the user, and **completing** the use case of Svach Servor 1.0.

Frontend

```
const workout = document.getElementById("workout").value.trim();
const meal = document.getElementById("meal").value.trim();
const cuisine = document.getElementById("cuisine").value.trim();
const ingredients = document.getElementById("ingredients").value.trim();
const scenario = document.getElementById("scenario").value.trim();
```

```
let userInput = `Today's Workout: ${workout}\nMeal of Day: $
let userShare = `I want to eat ${cuisine} food for ${meal}`;
```

```
const response = await fetch("http://127.0.0.1:5006/generate", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({
        prompt: userInput,
        sharePrompt: userShare
    })
});
```

```
const data = await response.json();
console.log("Received data:", data);
```

Backend

```
app = Flask(__name__)
CORS(app)

@app.route("/generate", methods=["POST"])
def generate():
    data = request.json  # Get user input from frontend
    user_prompt = data.get("prompt", "")
    user_share = data.get("sharePrompt", "")
```

```
top_links = meals_data.head(3)['video_url'].tolist()
link1 = top_links[0][17:]
link2 = top_links[1][17:]
link3 = top_links[2][17:]

return jsonify({
    "response": response.choices[0].message.content,
    "video1": link1,
    "video2": link2,
    "video3": link3
})
```

Strand (iii) Follow the plan to create the solution, which functions as intended.

The final digital solution is highly confluent with the initial solution for the problem that I had identified earlier, which was the lack of simultaneously palatable, cost-effective, personalized, **adaptive**, and accessible nutritional solutions for teenage school students. This will serve as a novel and innovative agent for the target audience of teenage school students, giving them the opportunity to find a solution like no other, and more importantly, one which is of higher quality than most of the alternatives currently in place. Before I developed this solution, the main characteristics of the problem which I identified involved how most of the current nutritional solutions for teenage school students were not **engaging**, not **immersive**, not **simple**, and resulted in teenagers not being **informed** about the diets they consumed. Furthermore, from my initial research, other solutions like recipe articles were found to be rather ineffective as they weren't **personalized** to the requirements of the users and also took a **long time** to read, being **unappealing** to them. Finally, my exploration of Artificial Intelligence-based chatbots such as ChatGPT, Gemini, and Perplexity made me realise that the models were rather **generic**, also having the chance for **hallucination**, producing and outputting **irrelevant** or **inconsistent** content. I kept these key pointers in mind while designing and developing my product, and the final outcome helped in reflecting key results, which helped in achieving the set objectives. I worked on making the product engaging, simple, and immersive, through two main methods, the first one was the entirety of the community page, gamifying the process of implementing **healthy practices** for teenage school students to achieve their nutritional requirements, as well as the **seamless** framework used in the product page, in order to assist the users with receiving **relevant** dietary recommendations. Next, I worked on making it personalised as well as appealing to the users mainly in the product page, in which I allowed them to receive **real-time** diet and recipe recommendations, based on a series of detailed and imperative **parameters** inputted by the user for their current circumstances and preferences. Finally, the implementation of **Retrieval-Augmented Generation** helped in **extending the knowledge** of the initial **Large Language Model API**, to ensure it outputted relevant, suitable, as well as reliable information to the users. Together, with the identified problems of the previous alternative solutions, and the strides that my product makes to mitigate them, I was able to develop an **effective** digital solution, a website called **Svach**, presenting a product known as **Svach Servor 1.0**, **designing healthy practices** which together create **significant** and **optimal** impact on **well-being**.

To create the final product, I mostly implemented the steps that I had outlined in the **initial** plan. I started by **gathering** and **creating** required images and icons, using sources such as Flaticon, RemoveBG, Google Images, and Google Docs, taking advantage of all the advanced features the tools had to offer, and synthesizing them to obtain the images I required. This culminated with the creation of a logo from my website, using the Samarkan font style to signify a sense of traditionality and impact. Next, in order to ensure I achieved the **Navigation** aspect of my success criteria, I worked on **creating** and **styling** a **navigation bar**, which allows the user to effectively traverse through the five pages of the website. I placed a logo at the extreme left, which represents the website, as well as serves as a link to the home page, along with links to all the webpages at the right side of the bar. I also effectively used differences in colour to help in informing the user which page of the site they are currently on, with the link for the current page being in white, and the rest of the pages being in beige, allowing it to serve as a simple and seamless mechanism for navigating through



Fig: Final Navigation Bar with Logo

the website. I then worked on **developing** the full functionalities for the Home, About, Community, and Contact Pages, along with the **frontend** for the Product Page.



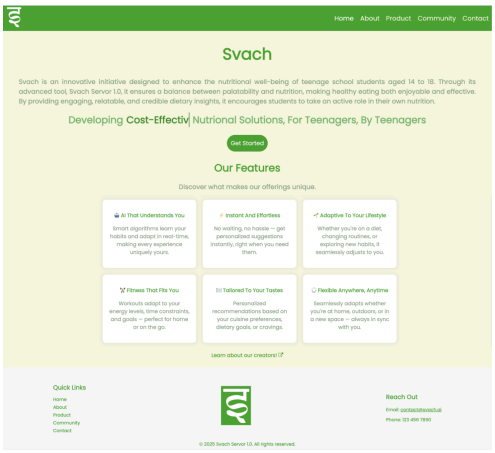
Fig: Information on Channels Used

Next, I worked on **building** a recipe dataset using only short-form videos, with a maximum duration of **five minutes**, mainly to ensure that the low average attention span of teenage school students wouldn't limit them from accessing and utilizing the features of the digital solution. More importantly, to ensure that I guaranteed a sense of ethos and credibility for the users, I strived to utilize videos only from reliable content creators, which I distinguished by relying on Government Bodies, Healthcare Providers, Certified Dietitians, and Channels with upwards of 100,000 Subscribers. This led me to refer to creators such as MOMables, Mayo Clinic, Mind over Much, and MyPlate by the United States Department of Agriculture.

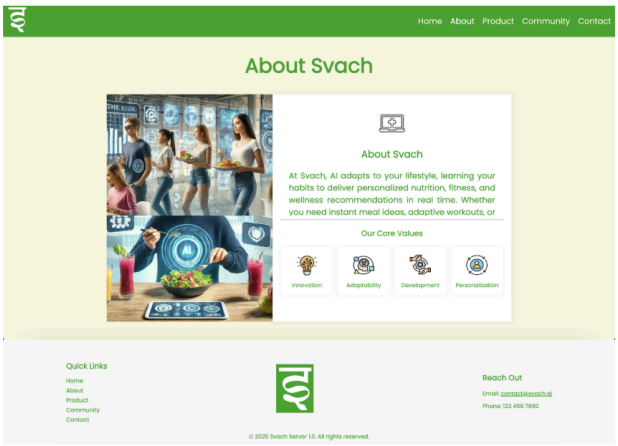
In order to receive relevant data to input to the Large Language Model, I needed to obtain relevant information from my dataset of recipe videos, for which I **extracted** the metadata and transcripts for each recipe video using Youtube Data API, getting enough text data to refine the model. I followers this by working on the **backend** and **integration** features of the product page, by **developing** the Retrieval-Augmented Generation-based Large Language Model using OpenAI API and Python, I then **implemented** the Flask Server to receive and transport user inputs from the fronted to the backend, and the model outs from the backend to the frontend, following which I **integrated** all the interactive and highly-technical features to the page, in order to **complete** its full functionality. However, a factor which I **didn't account for** in my initial plan was the extra time taken for **debugging** and **testing** the outputs of my code. There were numerous **syntax** as well as **logical errors** that I had encountered, which weren't explicitly expected or planned for earlier in Criterion C Stand (i), resulting in a deviation from the original action plan and the intended duration.

Developing Web Pages

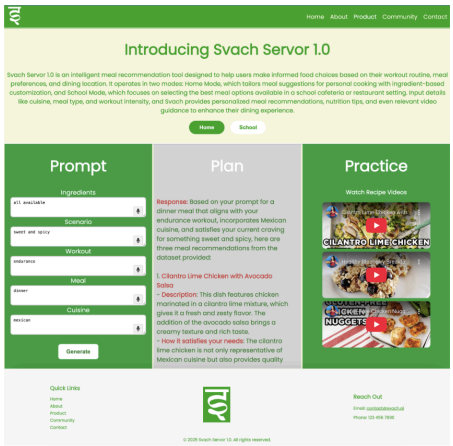
Home Page:



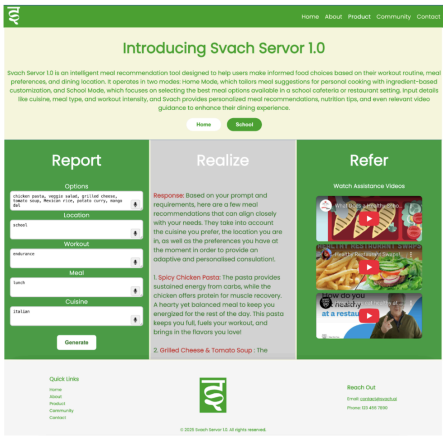
About Page:



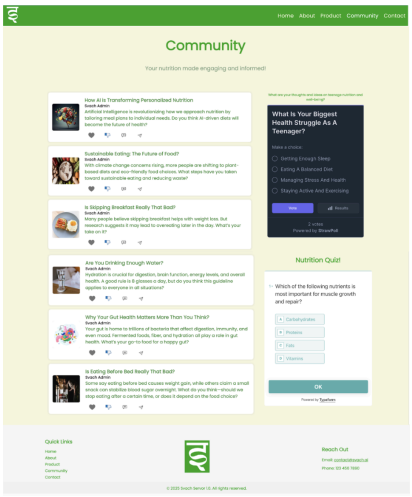
Product Page (Home Mode):



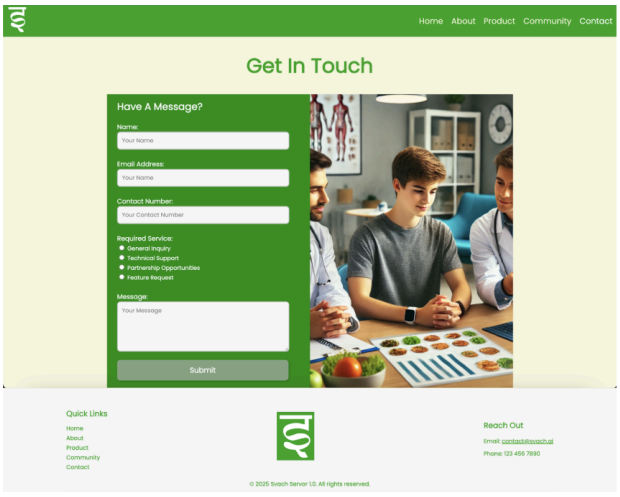
Product Page (School Mode):



Community Page:



Contact Page:



Strand (iv) Explain changes made to the chosen design and the plan while making the solution.

During the process of executing the design ideas and developing the digital solution, I had made numerous conscious choices to alter the initial plan, all for improving the final outcome of the solution with respect to its performance against the success criteria, solve the identified problem to a greater extent, or work upon unexpected problems. The changes made and the justifications for the same are listed below:

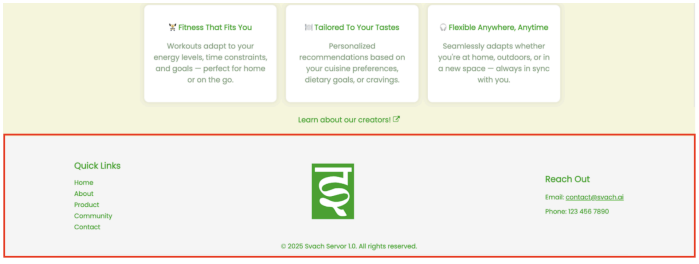
Change 1: Addition of Footer

Justification: While designing and developing my product, I referred to numerous other exemplar websites which I had interacted with before, and tried to take inspiration with regards to the intuitiveness and user-friendliness. While working on the development of the home page and analysing whether any additional components were required, I realized that a common and standard component of a majority, if not all of the websites was a basic, yet highly-impactful footer. This prompted me to add a footer to the end of all the five pages of my website, providing the user with quick links to access any webpage of the site, key contact information, an aesthetic display of the logo, and most importantly, a copyright message. Together, all of these components helped in making the digital solution more effective in serving as a **one-stop-shop** for teenage school students to develop **healthy practices** through **nutrition**, by ensuring that the entire interface and experience is seamless. Finally, it also allows in providing the users with any additional information they require, along with protecting the exclusivity of my original work, all in an aesthetic manner by smartly utilizing available space.

Initial Plan



Final Output



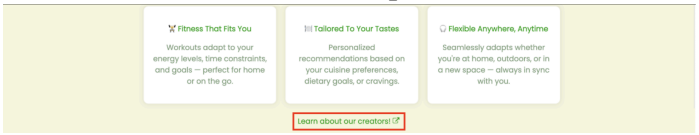
Change 2: Addition of External Link in Home Page for Outlining Content Sources

Justification: Throughout the process of researching the problem area, ideating an effective solution for it, and outline a success criteria to test it, one of the biggest parameters that I identified as relevant was providing the users a sense of credibility, with the content involved in the website. I wanted this to be unlike the regular short-form social media content consumed by teenage school students, which would be rather unpredictable and inconsistent, and thereby worked to ensure all the content I embedded was from certified or reliable sources. While this was part of the original design, I knew that I needed to gain as much trust and attraction from the audience as possible in order to ensure customer satisfaction, and more importantly, retention. That is why, I felt that explicitly sharing an external link to another document outlining the sources of the content, along with their origins, would help in establishing this trust, ensuring that my potential users would indeed be satisfied and impressed by the final outcome.

Initial Plan



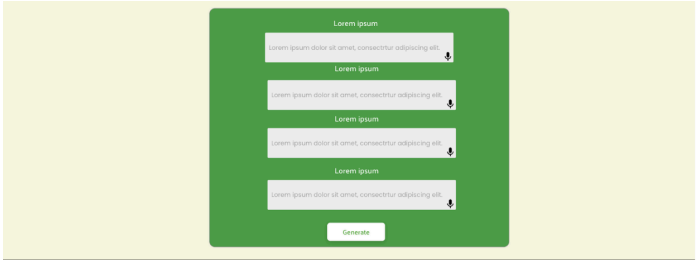
Final Output



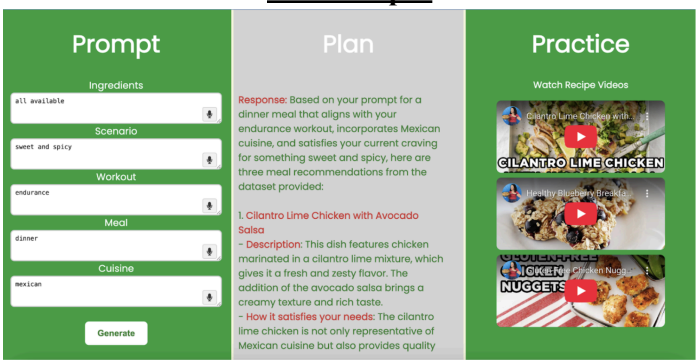
Change 3: Complete Modification of Product Page For Increased Simplicity

Justification: This change, in my opinion, helped me in completely redesigning the product webpage of my website, also the most important one of the five I had planned, and laid the foundation for a more successful and effective final outcome. My earlier workflow involved the user inputting their preferences through a form and submitting it, followed by an overlay on the same page giving them their dietary recommendation. However, when I was building the functionality of this webpage, I realised that using an overlay would be incredibly ineffective, as it would not only be temporarily active for the users to interact with, but also incredibly cumbersome to use, unintuitive, as well as highly problematical, due to the fact that it didn't help in providing immersive, informative, and innovative outputs to the user,
completely diverging from the very characteristics that I intended to include. This led me to completely revamp this page, eliminating the need for an overlay, and presenting all the required information on the page itself. I achieved this by dividing the whole page into three small sections, with the first on being 'Prompt', for the users to input their current preferences, availability, and scenario, the second one being 'Plan', for the users to receive a simple, yet highly helpful, plan from the output of the Retrieval Augmented Generation-based Large Language Model, and the final one being 'Practice', providing the users with the actual videos from the identified reliable content creators, with them matching the recipes mentioned in the plan provided earlier. This framework was the one for the Home mode, with a slight change for the school mode, which was 'Report', 'Realize', and 'Refer', following a similar format as the Home mode, but being more personalised for applications outside one's home, wherein they have lesser control over their options. This way, I converted my initial, less effective design, into a simple and seamless framework that any user could use in order to receive personalized, cost-effective, and quick dietary and recipe recommendations and plans.

Initial Plan



Final Output



Change 4: Utilizing CORS and Python In Place of Jupyter Notebook for Running Flask Server Backend

Justification: My initial plan involved implementing the Flask Server using just a Jupyter Notebook, due to its interactive nature and step-by-step workflow. However, when I was implementing and running the notebook, I ran into numerous errors, resulting in the code not being executed, because the process of running the Jupyter Notebook took up the required Python ports, which conflicted with the ones used by the server. Naturally, this led me to move from the Jupyter Notebook, and directly employ a Python file to avoid the overlapping of ports, which I achieved by converting the notebook to a script. Furthermore, when I was trying to run the Python file and researching online for similar implementations and industry standards, I came across a mechanism called CORS (Cross-Origin Resource Sharing), which I then implemented. Due to the fact that I was interacting with numerous web pages, domains, resources, and APIs during the development, I needed to implement a procedure which allowed me to access and control all the resources from these different origins, making CORS the optimal option, with its secure nature also influencing this decision. I didn't account for these two dynamics in my original plan, and implementing these changes resulted in me using different techniques than what I expected, but ultimately they helped in achieving my set goal.

Initial Plan

Implement Flask Server to receive user inputs from the Frontend of the Product page.	In order to transport the data from the frontend user input form, run it through the Retrieval-Augmented Generation-based Large Language Model, and revert back by displaying the meal and recipe recommendations to the user, a Flask server will be required to connect the frontend to the backend.	Visual Studio Code, Flask, Python, Jupyter Notebook, OpenAI API, JS, W3 Schools, Google Chrome	1.5

Final Output

```
from flask import Flask, request, jsonify
from flask_cors import CORS

app = Flask(__name__)
CORS(app)
```

```
product.py > ...
1   #!/usr/bin/env python
2   # coding: utf-8
3
    Run Cell | Run Below | Debug Cell
```

```
jupyter nbconvert --to script product.ipynb
```

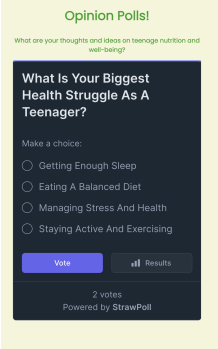
Change 5: Reduced Opinion Poll Embeds in Community Page

Justification: At the start, my final wireframe for the Community Page involved one nutrition quiz and two opinion polls, which the users could interact with in order to make the process of achieving their nutritional goals more informed and engaging for them. However, as I was developing the product, I realised that in order to create a sense of engagement, especially with the user group being teenage school students, anticipation was necessary. That changed my vision for this page, from just a one-time poll or quiz, to weekly polls and quizzes updated by the administrator, allowing to facilitate a scenario for the users in which after completing a quiz or a poll, they would eagerly be awaiting the one released the next week, allowing to increase customer retention, but more importantly, allow the users to invest more amounts of time and effort into their well-being. That is why, I changed the design from two one-time polls, to one weekly-changing poll, allowing to foster the sense of anticipation desired, and promote the various positive benefits that I had identified.

Initial Plan



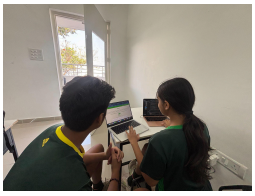
Final Output

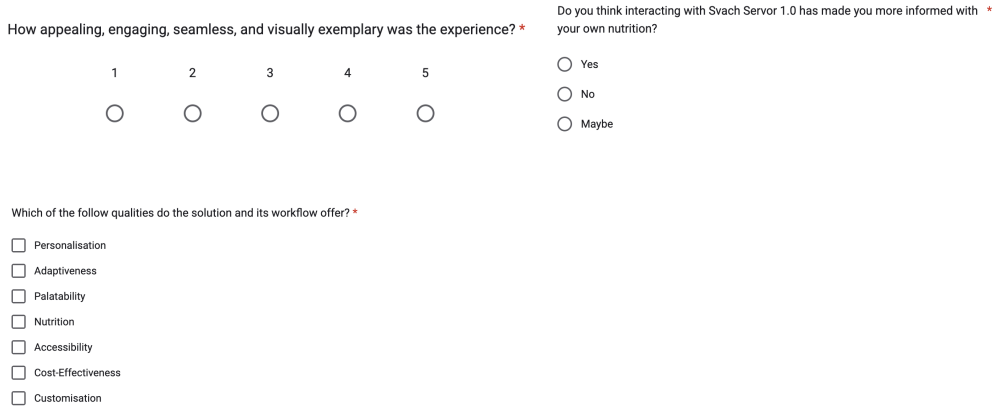
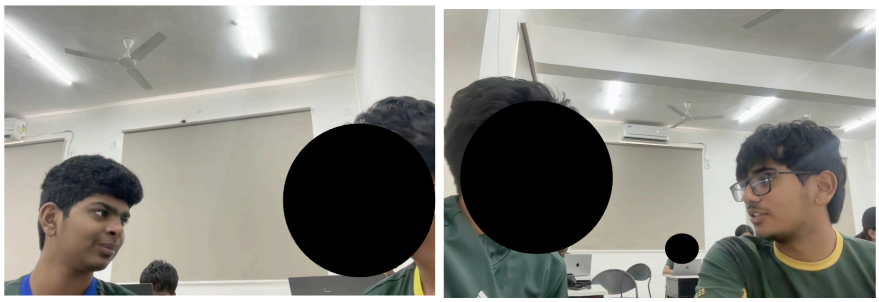


Criterion D - Evaluating

Strand (i) Design detailed and relevant testing methods, which generate accurate data, to measure the success of the solution.

In order to evaluate whether my **development** process and final product were effective in solving the identified problem, I comprehensively conducted the robust testing methods which I outlined earlier. **Detailed information and results from the tests can be found in Appendix E**.

<table>
<tr><td><u>Method</u></td><td colspan="2">1. Beta Testing - Peer White Box and Black Box Evaluations</td></tr>
<tr><td><u>Relevant Success Criterion</u></td><td colspan="2">Functionality, Accessibility, Navigation</td></tr>
<tr><td><u>Justification</u></td><td colspan="2">Before officially launching the product for full-time use by the target audience of teenage school students aged 14 to 18, it is imperative that a select, highly-knowledgeable, and exclusive section of it has the opportunity to experiment with its various features and possibilities to provide unfiltered and real-world feedback, along with key realisations on any potential bugs or errors. Through such an approach, featuring a blend between White Box and Black Box testing by the user group, I will also be able to obtain an honest as well as reliable understanding of where my digital solution and the product of Svach Servor 1.0 really stand.</td></tr>
<tr><td><u>Relevant Individuals and Tools</u></td><td>MYP Year 5 Students:
1. Mr. Aagam Jain
2. Ms. Aanya Pande
3. Ms. Saanvi Parige</td><td><u>Validity:</u> Technology-enabled MYP students will be highly skilled and experienced with utilizing web and Artificial Intelligence applications, and can hence help in effectively assessing both the internal and external environments of the digital solution through a qualified lens, while serving as members of the target audience, further allowing a fruitful investigation.</td></tr>
<tr><td><u>Evidence</u></td><td colspan="2">Black Box Evaluations - Functionality and Accessibility:
  
White Box Evaluation - Navigation:
  </td></tr>
</table>

Method	2. Acceptance Testing - Target Audience Interviews and Survey	
Relevant Success Criterion	Customer, Aesthetics	
Justification	For any product, acceptance testing is imperative as it will help in validating whether or not it actually achieved its purpose of solving the identified problem faced by the target audience. Through this method, I will be able to directly obtain the opinions of the user group of **teenage school students** on the efficacy of the digital solution, as well as make key conclusions on key metrics such as the **User Interface (UI)** and **Aesthetics**, **User Experience** and **Functionality**, and most importantly, **Customer Satisfaction** and **Performance.**	
Relevant Individuals and Tools	Target Audience Members: 1. MYP Years 4-5 Students (10) 2. Mr. Saketh Kollipara 3. Mr. Rohit Raju	**Validity:** Students of MYP Years 4-5 will fall under the exact target audience of teenage school students aged 14 to 18, making them optimal. Also, Mr. Saketh Kollipara and Mr. Rohit Raju are the same students I interviewed earlier for researching the problem, and examining their opinions on the digital solution will allow me to truly understand its profound impacts.
Evidence	Survey Questions:  Interviews: 	

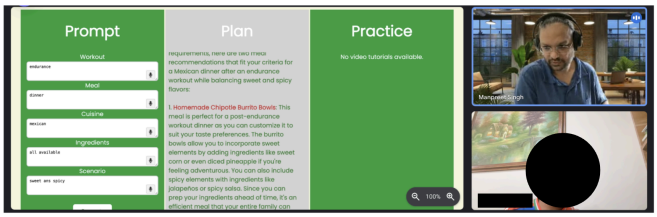
Method	3. Accounting Sheets
Relevant Success Criterion	Cost

Justification	Even though this aspect may not seem as directly related to the success of the product and its development product, retrospectively analysing the **costs** involved in developing the digital solution will help in understanding how **productively** I had used my available resources to fulfill my preset goals. Utilizing accounting sheets will assist me in representing the total costs incurred in a **visual** and **interpretable** manner. Ensuring that overall monetary and time costs were indeed minimal will allow to launch the product for free-of-cost as there will be barely any capital and effort to recover or compensate, also making the process of creating the solution much more **effective** and **accurate**.	
Relevant Individuals and Tools	Data Table Logging and Calculation Software: 1. Google Sheets	**Validity:** Google Sheets is an industry-renowned tool, which can be used to seamlessly create spreadsheets for presenting, transforming, and organizing data, all of which will be relevant when tracking the time and money spent on the development of the product.
Evidence	Monetary Cost: Time Cost:	

Item	Cost (USD)
Visual Studio Code	-
Frontend HTML Framework	-
Frontend CSS Framework	-
Frontend JS Framework	-
Backend Python Flask Server	-
Jupyter Notebook	-
Youtube Recipe Playlists Datasets	-
Pandas, Numpy, OS Libraries	-
Youtube Data Developer API	-
OpenAI Developer API	10
Total Cost: $10	

Task	Time (Hours)
Gathering Content	0.5
Styling Navigation Bar	0.5
Developing Home Page	2
Developing About Page	0.5
Developing Community Page	2
Developing Contact Page	1
Developing Product Page Frontend	1.5
Collect Recipe Dataset	1
Extract Metadata and Transcripts	0.5
Develop RAG-based LLM Framework	3
Implement Flask Backend	3
Integrate Interactive Features	1
Complete Product Page	2.5
Debugging and Refining	3
Total Duration:	22

Method	4. Expert Evaluation - AI Engineer Appraisal	
Relevant Success Criterion	Safety	
Justification	This test was one of the most crucial ones, due to the fact that it involved interacting with an industry professional, someone highly skilled in the domains of Artificial Intelligence (AI) and Machine Learning (ML). My consultation helped in not only gaining insights primarily on the overall **safety** and **privacy** of my digital solution, but also resulted in me obtaining numerous other inputs to understand the strengths as well as required improvements for the **functionality** of the product.	
Relevant Individuals and Tools	Subject Expert: 1. Mr. Manpreet Singh - Senior Director, Software Engineering and AIOps, Salesforce	**Validity:** Mr Singh has acquired numerous degrees, written books, as well as developed various projects in the AI and ML community, making him a highly-qualified and experienced individual to evaluate the safety of my product.

Evidence	

Method	5. Compatibility Testing - Different Devices and Browsers	
Relevant Success Criterion	Accessibility	
Justification	Ensuring that the final outcome of my website would be compatible with different devices and browsers was important, as a potential user could use any combination of devices and browsers according to their own preferences, making it important they can still effectively utilize and implement the digital solution, as well as the product of Svach Servor 1.0, in whatever scenario and setup they wish, assessing the overall sense of **accessibility**.	
Relevant Individuals and Tools	Device Manipulation: 1. Google Chrome DevTools Browser Manipulation: 1. Google Chrome 2. Apple Safari 3. Mozilla Firefox 4. Microsoft Edge	**Validity:** All the four browsers utilized for testing are industry standard and are some of the most common web browsers used. Meanwhile, the DevTools framework will allow to adjust the HTML content directly from the browser, ensuring that it is adequately and effectively represented in various device formats. Together both these tools will ensure that the product is evaluated based on a vast range of increments and scenarios.
Evidence	Devices: Dimensions: Macbook Pro 14" Dimensions: iPad Pro Dimensions: iPhone 14 Pro Max Browsers: Chrome Firefox Safari Edge	

Strand (ii) Evaluate the success of the solution against the design specification.

After design as well as implemented comprehensive testing methods for my digital solution, I was able to obtain eye-opening as well as invaluable insights and information on its efficacy in tackling the identified problem, meeting the requirements of the design specification, as well as satisfying the target audience of teenage school students, both in a quantitative as well as qualitative manner. Furthermore, I was able to ensure that all the data I had gathered was reliable, relevant, and informative, mainly be exploring various numerical, mathematical, and visualization based techniques to better comprehend the quantitative insights, while only relying on the sense of credibility offered by experts, subject-matter experts, peers and facilitators for the qualitative insights, to ensure a sense of validity and value, despite a limited sample size. Through this process, I transitioned from my earlier partial and limited evaluation, to the current holistic and completely informed evaluation. Retrospectively, that initial assessment was rather inflated due to the lack of proper insights on the actual functioning and field implementation of the product, but through this evaluation, it became more concrete, and I was more aware of where exactly the digital solution stood. Furthermore, it gave me a better understanding of how I could

improve the digital solution, which can be implemented in order to maximise the strong potential that it already has made strong progress towards. What made these understandings more conclusive was the fact that they were all derived from the credible and valid sources that I outlined above, instilling a sense of confidence in me that my evaluations and reflections were as accurate as possible.

Evaluation of Digital Solution based on Analysis of Testing Methods:

Criterion	Specifications	Score	
Aesthetics	The digital solution must feature a warm, geometric, sans-serif font style.	2/2	8/8
	Clear as well as informative graphics and images throughout the pages of the digital solution.	2/2	
	The digital solution must use a light, calming, and balanced colour scheme.	2/2	
	Overall minimal and sophisticated appearance of the final digital solution.	2/2	
Cost	Monetary costs as a part of the developmental process should only be limited to purchasing necessary API keys, hosting and deployment services, and domain names.	2/2	7/8
	The total monetary cost incurred cannot exceed 20 United States Dollars.	2/2	
	Time costs as a part of the developmental process should only be limited to ideating and researching the technical aspects required for the digital solution, programming the digital solution, and testing its performance.	2/2	
	The total time cost incurred cannot exceed 20 hours.	1/2	
Customer	The digital solution must cater to the predetermined demands of the determined target audience of teenage school students, within the age range of 14 to 18 years.	2/2	8/8
	The provided content as well as structure of the digital solution must offer a sense of credibility and reliability, by ensuring that it is certified, relatively short in duration, and highly-engaging.	4/4	
	Offers a simple and highly effective User Interface, avoiding being cumbersome and complicated.	2/2	
Functionality	The digital solution's tasks need to involve making effectual, seamless, informed, and cost-effective meal, recipe, and diet recommendations of palatable nutritional solutions to teenage school students.	1/2	7/8
	The recommendations and generations provided by the digital solution must be based on the personalised, spontaneous, and adaptive calibration of key data points, including available ingredients or meal options, intensity level of physical activity completed or expected, along with current mood, cuisine and dietary preferences.	3/3	
	The program of the digital solution must be powered by an Artificial Intelligence-based workflow which assists in the dynamic recommendation and generation of meals and recipes for teenage school students.	3/3	
Accessibility	The digital solution must be available to the user group of teenage school students free-of-cost, due to their financial dependence.	2/2	6/8
	The program of the digital solution needs to be implementable in versatile setups, including one's home, their school, or a restaurant, by having different features which are adaptive and personalized to the users' unique circumstances and requirements.	2/3	

	The final digital solution has to be effectively accessed in numerous devices such as laptops, mobile phones, and tablets, as well as different browsers such as Google Chrome, Safari, Microsoft Edge, and Mozilla Firefox.	2/3	
Safety	The digital solution must not access or store any Personally Identifiable Information (PII) or Health Data of the users, except their full name, email address, or phone number, only when required.	3/3	7/8
	The digital solution must not output inconsistent, inaccurate, or non-scientific information and advice to the target audience group. Any Artificial Intelligence-based systems cannot be implemented using general training data, in order to prevent hallucination.	2/2	
	Any information or data that needs to be stored or transported will be done only using secure, reliable, and industry-acclaimed secure servers or protocols.	1/2	
	Must ensure that any Large Language Model API employed will not implement the user's inputs for training processes.	1/1	
Navigation	The final digital solution must have no less than 4 and no more than 6 main pages in order to ensure simultaneous elaboration and conciseness.	4/4	8/8
	Features an understandable as well as clear and organised standout navigation/menu bar.	1/1	
	The digital solution must have straightforward and simple structure and navigation, allowing users to utilize every one of its aspects, while helping in avoiding confusion.	3/3	
Final Grade: Exceptional		51/56	

Justification of Evaluation:

Criterion	Final Score	Justification	Evidence
Aesthetics	8/8	The digital solution indeed features a warm, geometric, sans-serif font style, as well as informative graphics and images throughout. It uses a light, calming, and balanced colour scheme, which contributes to a minimal and sophisticated appearance.	Test 2 - Target Audience Survey
Cost	6/8	Monetary costs as a part of the developmental process only include purchasing necessary API keys, hosting and deployment services, and domain names, all of which come under 20 United States Dollars. The total time cost is indeed ideating and researching the technical aspects required for the digital solution, programming the digital solution, and testing its performance, but it doesn't come under 20 hours.	Test 3 - Accounting Sheets
Customer	7/8	The digital solution highly caters to the predetermined demands of the determined target audience of teenage school students, within the age range of 14 to 18 years, mostly providing highly-engaging content, which is credible and reliable, along with a seamless User Interface.	Test 2 - Target Audience Interviews
Functionality	7/8	Mostly effectual in making seamless, informed, and cost-effective meal, recipe, and diet recommendations of palatable nutritional solutions to teenage school students, which are highly personalised and spontaneous,	Test 1 - Black Box Beta Testing

		being based on data points inputted by the user, and the power of an Artificial Intelligence Workflow.	
Accessibility	5/8	The digital solution will be available to the user group of teenage school students free-of-cost, making sure that their financial dependence isn't a limitation. Can be implemented successfully in multiple browsers, as well as in home environments, but has scope for improvement with respect to smaller devices as well as in school or restaurant setups.	Test 1 - Black Box Beta Testing Test 5 - Compatibility Testing
Safety	6/8	The digital solution doesn't access or store any Personally Identifiable Information (PII) or Health Data of the users, except their full name, email address, or phone number, that too only when required particularly. All the information and advice outputted to the user is consistent, accurate, and scientific in nature, being specific and impactful, with none of the user's inputs being used for training the models of the API. However, the Flask Server and lack of the use of a security protocol for it make it a bit more vulnerable to rare cases of attacks or breaches.	Test 4 - AI Engineer Expert Appraisal
Navigation	8/8	The final digital solution has between 4 and 6 main pages, a clear and organised standard navigation bar, and an overall straightforward structure, in order to ensure simultaneous elaboration, clarity, and conciseness.	Test 1 - White Box Beta Testing

Strand (iii) Describe how the solution can be improved.

Based on the above testing methods conducted, it can be seen that the product was "**exceptional**" in its approach to solving the problem identified initially. However, there were still a few strands and aspects in which it wasn't able to perform to the fullest extent possible, with the results of the tests showcasing a few weaknesses to work upon, and thereby leaving substantial room for improvement, which are elaborated upon and explained below.

Weakness	**Occasional potential for ineffective diet and recipe recommendations**
Source of Identification	**Success Criteria - Functionality:** The functionality aspect of the success criteria requires the digital solution to provide effective and relevant dietary recommendation to teenage school students, which isn't observed in certain test cases. **Test 1 - Peer Black Box Testing:** During the process of testing the functionality of the digital solution and its product webpage, the various cases and scenarios tried by the MYP students resulted in consistently obtaining results in which a maximum of 2, and on average 1, out of all the three outputted recipes were only relevant to their inputs, making it less effective.
Improvement	The main step that can be taken to overcome this challenge should be to extend the recipe video dataset from the current size of around 200 videos, to about 350 to 400 videos, allowing to obtain a greater variety of videos that can potentially be outputted to the user. As a result, the Retrieval-Augmented Generation-based Large Language Model will be applied to identity as well as share more relevant recommendations to the users, in order to rectify this issue.

Weakness	**Possibility for vulnerability to data breaches, attacks, and exploits**
Source of Identification	**Success Criteria - Safety:** The safety aspect of the success criteria requires that the transmission of data within the digital solution would be done only through secure, reliable, and industry-standard servers, which has only

	been partially achieved. **Test 4 - AI Engineer Appraisal:** During my interaction with Mr. Singh, he shared an interesting point that even though it would be really rare, having incredibly minute to no chances of occurring, there would be a chance for individuals with malicious intentions to intercept the Flask Server employed and access the data being transported, making it less secure for the users.
Improvement	Fortunately, within the same discussion with Mr. Singh, he recommended exploring the implementation of mTLS (Mutual Transport Layer Security), which allows for improved encryption, authentication, as well as trust mechanisms within the digital solution. Consequently, it will ensure that the digital solution and the Flask Server will have enhanced security, being impervious to man-in-the-middle (MITM) attacks and unauthorized access.

Weakness	**Lack of content depth for the school mode in the Product Page**
Source of Identification	**Success Criteria - Accessibility:** The accessibility aspect of the success criteria mandates that the digital solution can be effectively implemented in a variety of locations and setups, both in the homes of teenage school students, as well as when eating outside, either in schools or restaurants. This balance isn't visible for the school mode, which isn't implemented as effectively as the home mode. **Test 1 - Peer White Box Testing:** After the MYP students examined both the internal as well as external characteristics of the digital solution, and especially the product page, they had identified that while the framework and interactions involved with the home page were helpful, similar qualities weren't completely expressed for the school mode, with the content provided not being in-depth, also lacking opportunities for further insights.
Improvement	A seamless measure which can be taken to address this issue will be working on two aspects of the school mode framework, **Realize** and **Refer**. For the former, rather than just having an output of the Large Language Model, I will directly integrate a similar Large Language Model-based chatbot, but one which allows the user to go back and forth with the Artificial Intelligence workflow, to clear any potential doubts, and ensure a sense of efficacy. Also, the content displayed for the Refer section can be updated regularly, rather than keeping it the same throughout, to foster the desired sense of engagement as well as anticipation, refining the overall quality and experience with this mode, and **maximizing performance**.

Weakness	**Incompatibility and diminished aesthetics with smaller device types**
Source of Identification	**Success Criteria - Accessibility:** The accessibility aspect of the success criteria also involves the digital solution being able to be applied and utilizes under different device, such as Laptops, Mobile Phones, and Tablets, as well as browser, such as Chrome, Safari, Firefox, and Edge, environments, with the product lacking with regards to the former - devices. **Test 5 - Compatibility Testing:** During the process of testing the compatibility of the website with the different browsers and devices, the main issues arose when the website was used in mobile phone as well as tablet settings, in which certain containers and overlays took up excess space, while others had a shortage, altogether prompting to an interface with aesthetics of lesser effectiveness.
Improvement	There are a series of different solutions which I can implement to work upon this issue. They include using the HTML **meta tag**, along with **viewport and width attributes,** as well as modifying the CSS code to avoid direct pixel units, but rather **percentage units**, to ensure that the styling of all the containers will be **relative** to the device used. Finally, I will also implement CSS **media queries** in order to adjust the layouts for the varying screen sizes, ensuring a sense of **adaptability**, which will synergise with all the other measure to ensure the digital solutions is indeed compatible and aesthetically pleasing to the user, when used in other devices, especially smaller types.

Strand (iv) Describe the impact of the solution on the client/target audience.

Initial Problem Statement, Solution, and Global Context Connection

From the start of this process, all the way during Criterion A, the sole focus area was working around the problem of the lack of nutritional solutions for teenage school students balancing palatability, personalisation, and accessibility. My research helped me understand that the previously-existing solutions were barely effective in solving the problem, offering complex and cumbersome interfaces and experiences, being trivial and generic in nature, as well as too expensive for a teenager to afford. This led me to develop a product which satisfied all the aspects that the problem statement required to solve, mainly through the integration of an Artificial Intelligence-based workflow in the form of a Retrieval-Augmented Generation-based Large Language Model, allowing to generate credible, real-time, and cost-free dietary recommendations to the users. Finally, the entire product and its concept helped in establishing strong connections with the **global context** of **scientific and technical innovation**. Throughout the digital solution, I used my **understanding of scientific principles** in the implementation of the diet, meal, and recipe recommendations, and executed skills with **technical development** in order to ensure efficiency in solving the problem, ultimately unleashing the positive **impacts of scientific and technical innovations on communities and societies.**

Impact on the Target Audience

Through the various tests that I had carried out in Criterion D, Strand (i), I gained successful insights from most of them, indicating that the product was indeed **fruitful** in solving the identified problem, while **meeting the demands** of the target audience. Results from Test 1 (**Refer Appendix E.1**) indicated that the testing group comprising technologically-enabled MYP students, also part of the target audience, were majorly satisfied with the **Functionality, Accessibility,** and **Navigation** of the digital solution. Furthermore, the surveys and interviews for Test 2 (**Refer Appendix E.2 and E.3**) helped in validating that the product did have a positive impact on the target audience, with **83.3%** of the respondents express that it did help them make strides toward becoming immersed into their personal nutrition, along with **100%** of them rating the aesthetics either 4 or 5, on a scale of 5, along with reliable, direct, and unfiltered information from two members of the target audience, **Mr. Rohit Raju** and **Mr. Saketh Kollipara**, with both of them reaffirming that the product successfully satisfied their demands, while helping to work upon the same problems they shared during my research process of Criterion A, Strand (ii). Together, all of these tests helped in confirming that the product was **successfully accepted** by the members of the target audience, setting the scope for more widespread positive impact once it's launched in a mainstream fashion.

Fulfillment of the Design Brief and Drawbacks

Throughout the process of designing and developing the digital solution, the detailed and comprehensive **design brief** that I had composed in Criterion A, Strand (iv), along with the **design specification** and **success criteria** defined in Criterion B, Strand (i), served as constant sources of guidance, keeping me on the track of success, accuracy, and optimacy. I obtained a final score of **51/56**, meaning that the product was **Effective** in its approach to solving the problem, outperforming the earlier evaluation of the final design ideas in Criterion B, Strand (iv). However, the results of the tests conducted in Criterion D, Stand (i) left the scope to improve on its **Safety, Accessibility,** and **Functionality** features to maximize positive output, based on Criterion D, Strand (iii).

Impact on Self and Learning

After completing this E-Portfolio, I have experienced significant learning transformations, both in the technical as well as non-technical realms, powered by my understanding of **scientific and technical innovations**. I've throughout understood the scientific importance of a healthy and balanced diet, as well as the various nuances involved in achieving one, garnering greater motivation to pursue **healthy practices** myself. Moreover, I have been able to explore the intersection of two subject areas which I've simply adored for as long as I remember - **computer science** and **mathematics**. Most importantly, I was able to venture into a highly specific subdomain of my favorite interest area of Artificial Intelligence and Machine Learning, which was Retrieval-Augmented Generation **development**, one which I never explored before. I can now analyse problems better, **adapt** to a series of scenarios much more persistently due to the various errors and unexpected factors I encountered during the process, and finally, find newer and more impactful ways of implementing my interest in **science and technology** to solve the **world's biggest problems**. My journey of pursuing technology, computer science, and artificial intelligence has just begun, and this project has definitely propelled me to achieve unconventional success in the future, adding high amounts of fuel to my never-ending fire of a desire for innovation.

Refer Appendix F for ATL Skills Reflection